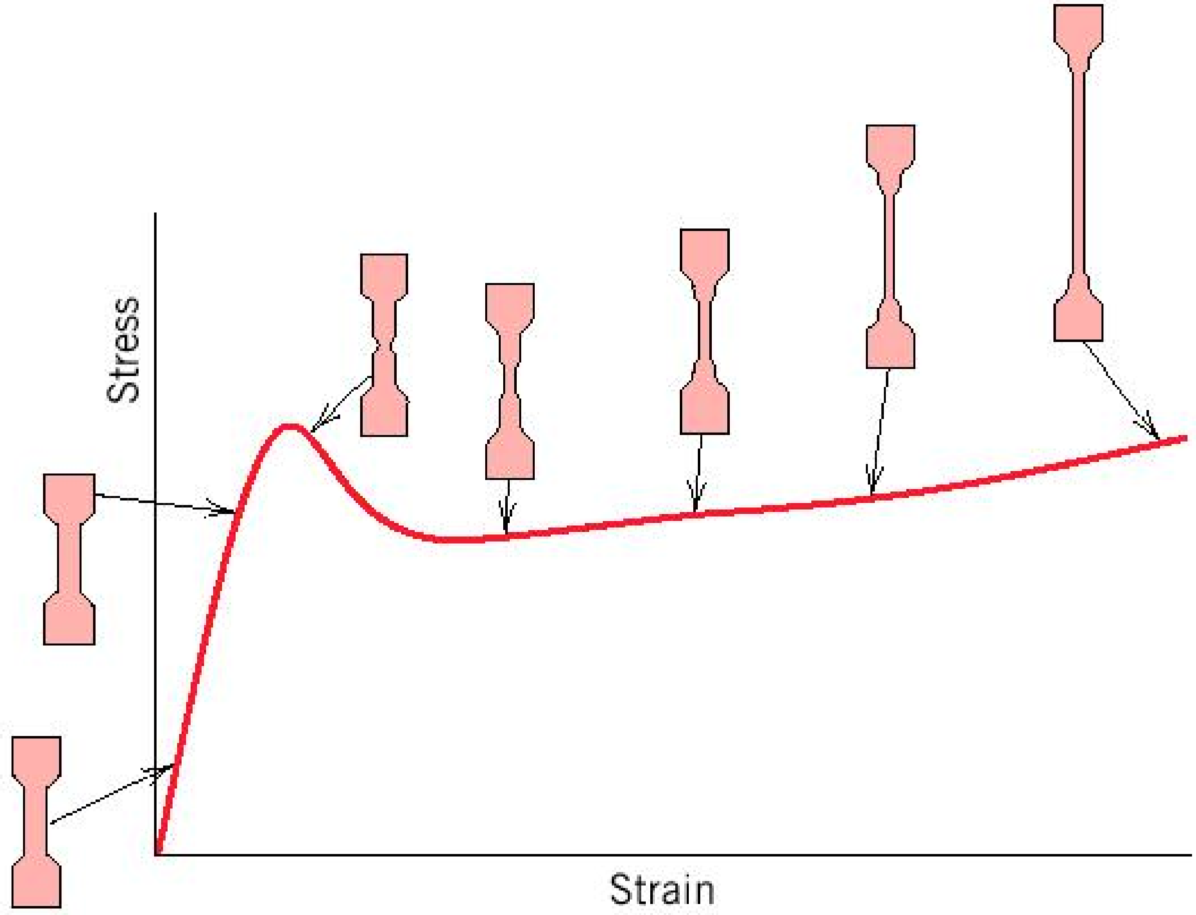
How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don’t cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)…**
- Ask questions and Think skeptically

Important Points from Last Lecture



Lecture 6:
Polymer Blends and Composites

SUN Dazhi

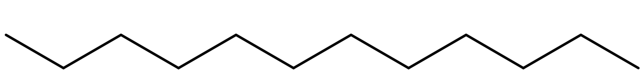
Polymer Blends: Polymer-Polymer Solutions

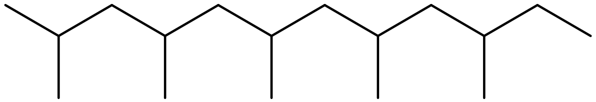
- **Miscible blends**: averaging of properties of two different materials. *Raising the T_g or strength or toughness of a cheap polymer.*

- **Immiscible Blends**: composite structures- properties may be synergistically greater than those of constituent materials

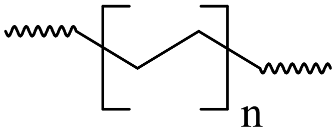
Polymer Solubility

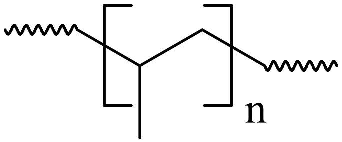
- **When two hydrocarbons such as dodecane and 2,4,6,8,10-pentamethyldodecane are combined, we generate a homogeneous solution:**





- **It is therefore interesting that polymeric analogues of these compounds, poly(ethylene) and poly(propylene) do not mix, but when combined, produce a dispersion of one material in the other. Why ?**





Blends: Thermodynamics

$$\frac{\Delta G^M}{RT} = \frac{\Phi_A}{N_A} \ln \Phi_A + \frac{\Phi_B}{N_B} \ln \Phi_B + \chi \Phi_A \Phi_B$$

$$\boxed{\chi_H = \frac{V_m}{RT}(\delta_A - \delta_B)^2}$$

where Φ = volume fraction
N = degree of polymerization
and χ = Flory's interaction parameter

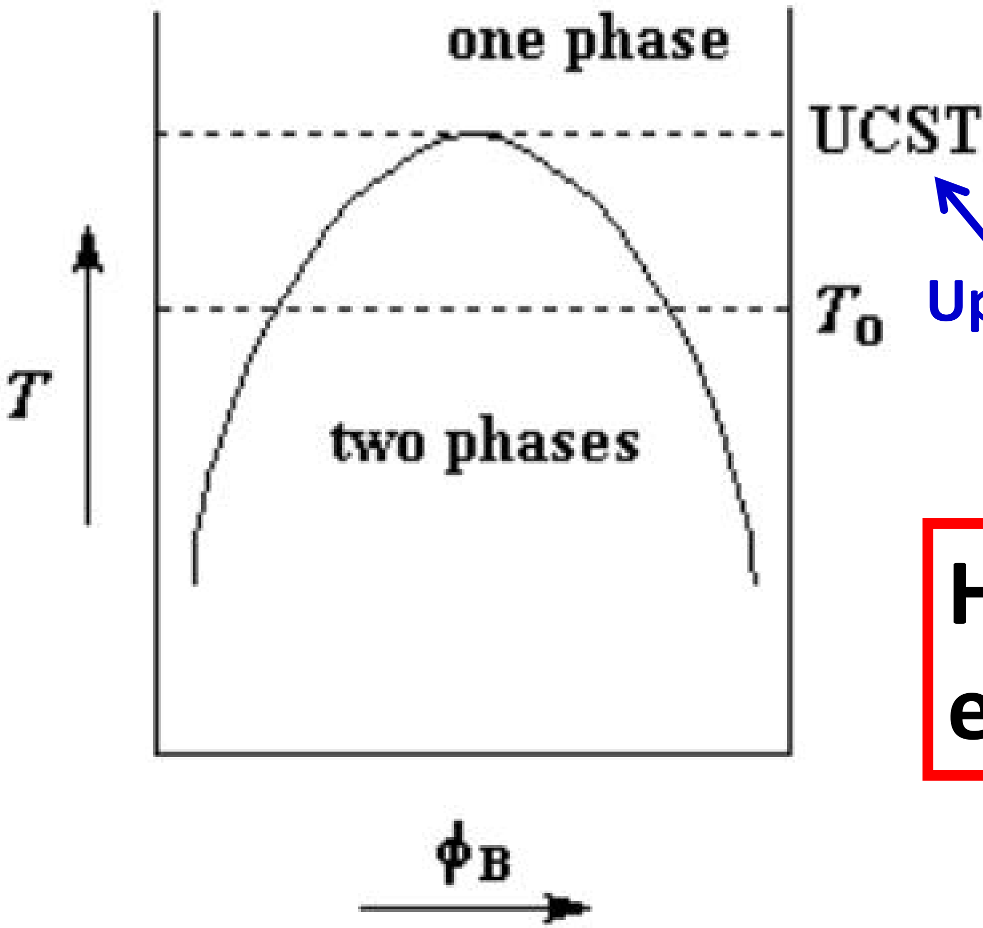
Entropic contribution (usually can be ignored)

Enthalpic difference in solubility: $\Delta\delta \geq 0$

- **Minimize difference in solubility parameters ($\Delta\delta = 0$)**
- **But even small positive $\Delta\delta$ will cause $\Delta G > 0$**

Polymer Miscibility Phase Diagrams

No strong non-bonding interactions & *theta solvent*

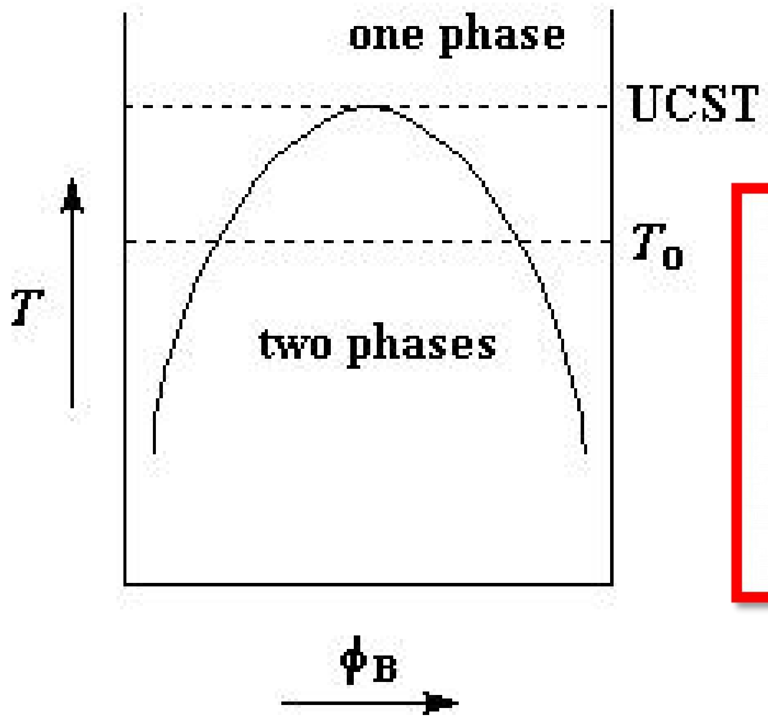


Upper critical solution temperature

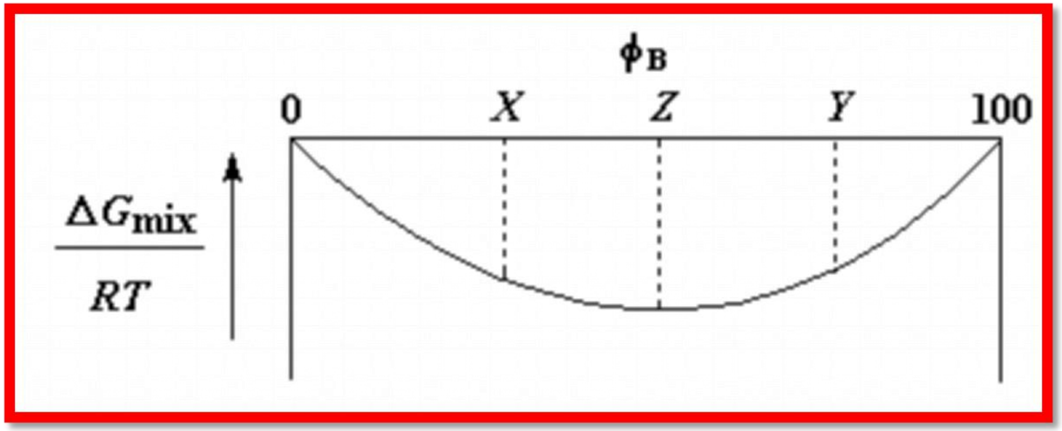
Heat up mixture & entropy rules

Polymer Miscibility Phase Diagrams

Free Energy

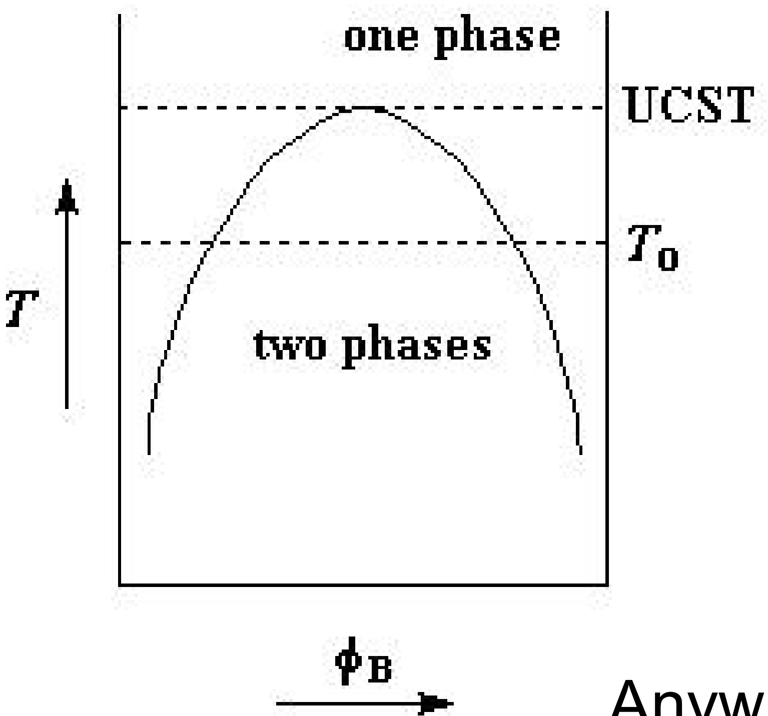


Single Phase Region

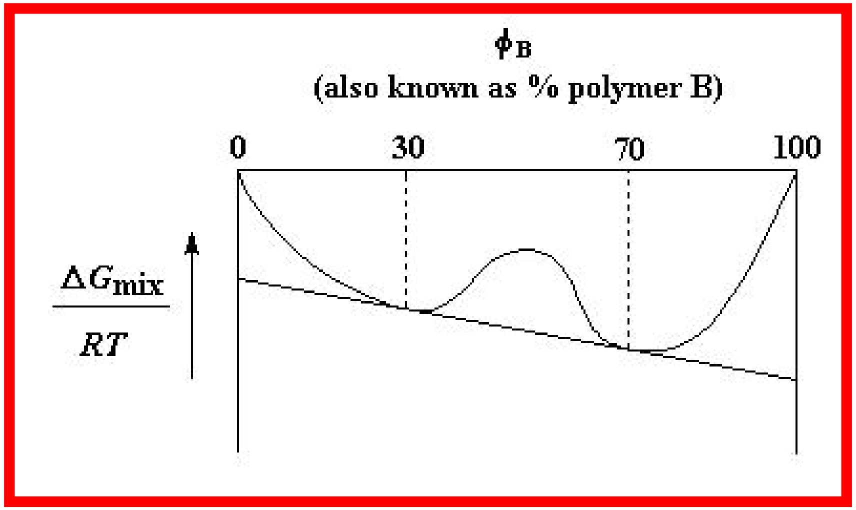


Polymer Miscibility Phase Diagrams

Free Energy

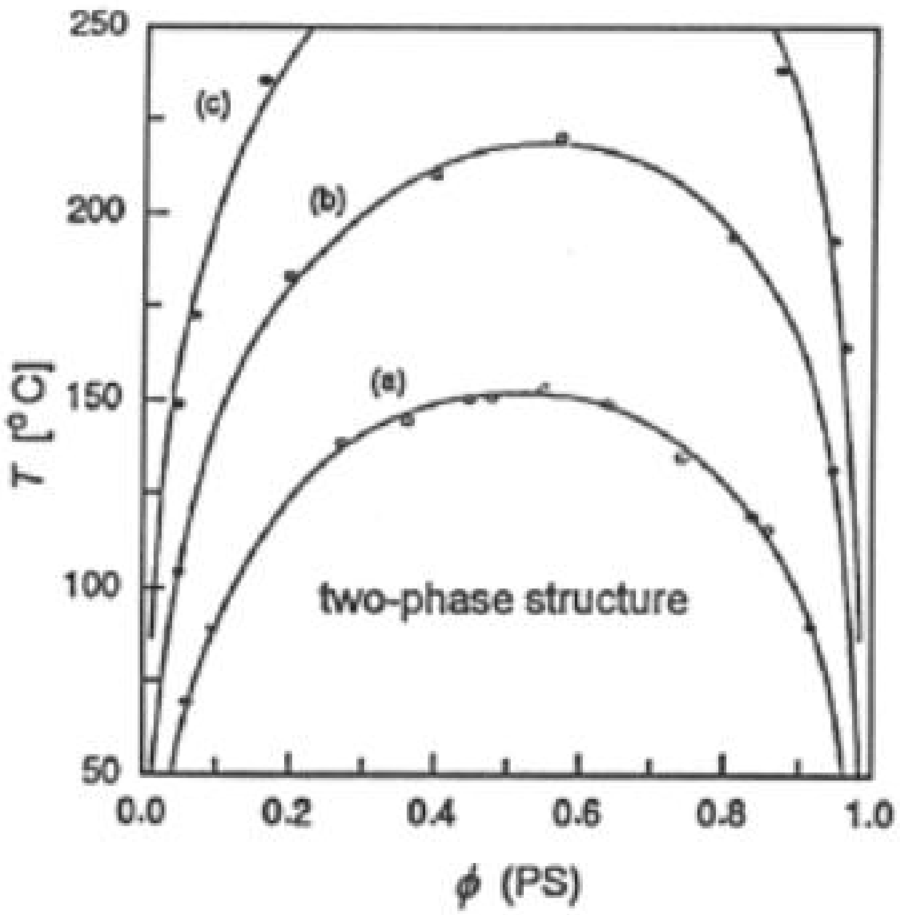


Two Phase Region



Anywhere between 30 & 70% leads to decomposition to two phases with 30 & 70% comps.

Effects of Mw On Phase Separation

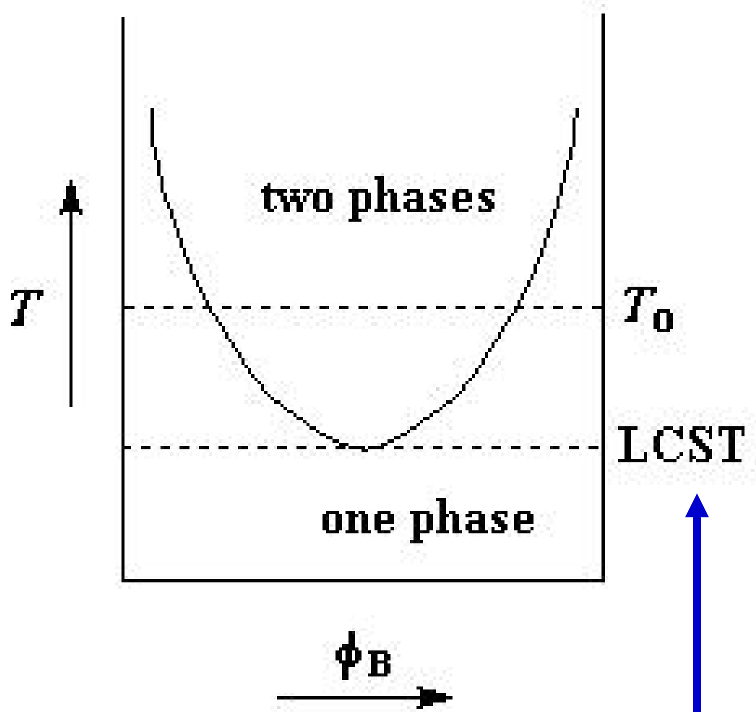


M(PB) = 2350

- M(PS) = 2250
- M(PS) = 3500
- M(PS) = 5200

Polymer Miscibility Phase Diagrams

Strong Non-bonding Interactions



A new term needs to be added to free energy equation to account for non-bonding interactions

$$\frac{\Delta G^M}{RT} = \frac{\Phi_A}{N_A}\ln \Phi_A + \frac{\Phi_B}{N_B}\ln \Phi_B + \chi\Phi_A\Phi_B + \frac{\Delta G_H}{RT}$$

where ΔG_H = free energy of hydrogen bond formation

$= f\,(\Phi_{A'}, \Phi_{B'}, V_{A'}, V_{B'}, K_{A'}, K_B)$

Lower critical solution temperature (LCST)

More Thermodynamics

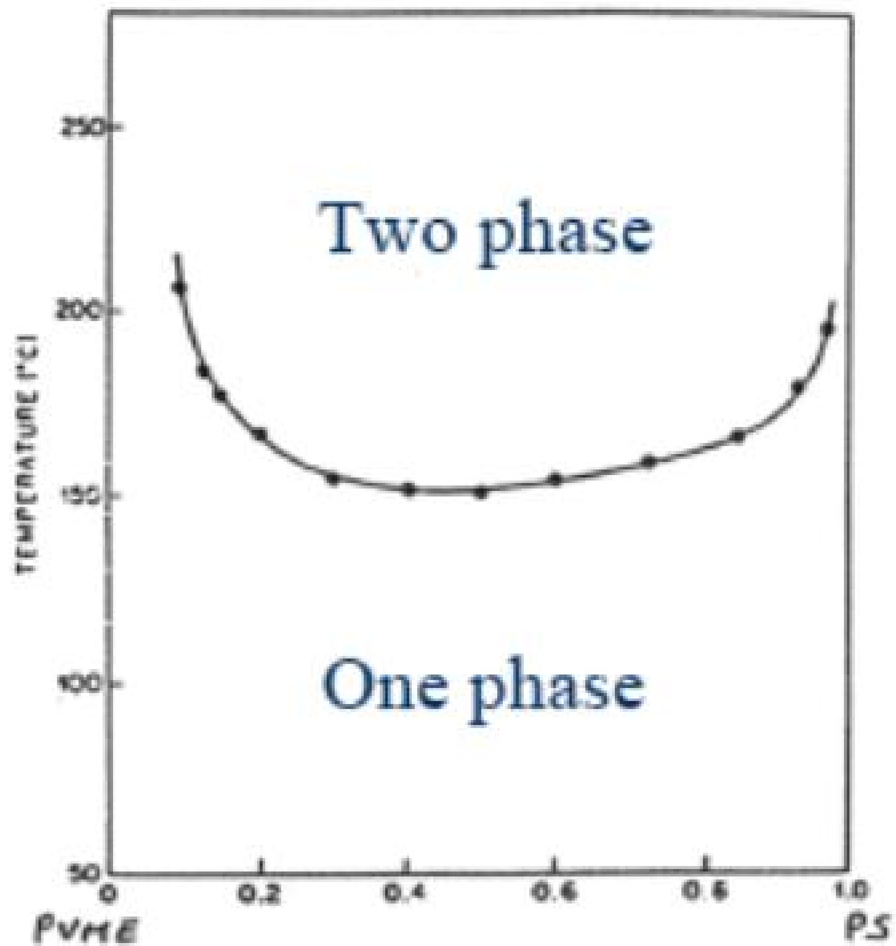
If we ignore the entropic contributions then the equation simplifies to:

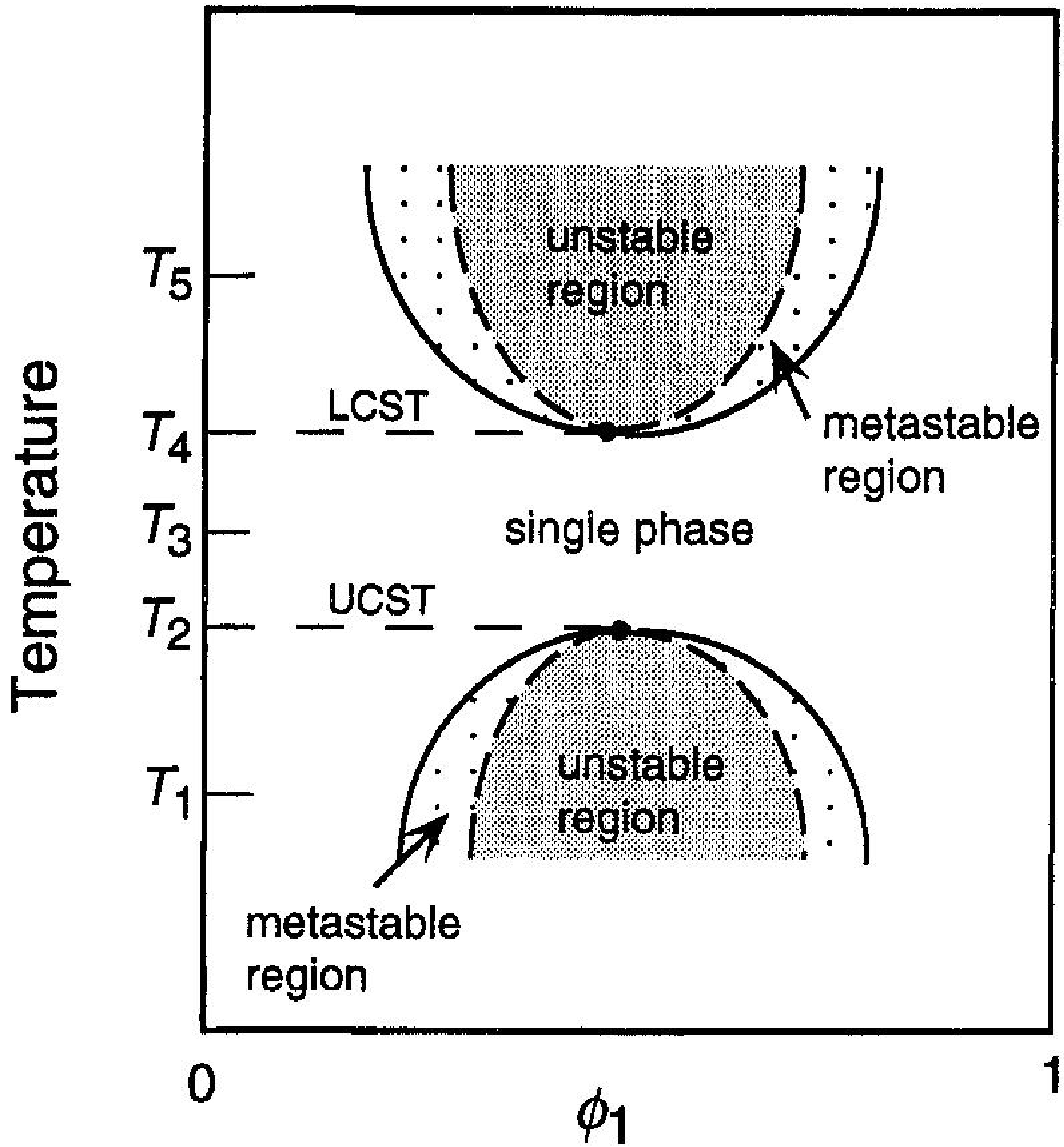
$$\frac{\Delta G^{M}}{RT} = \Phi_A \Phi_B \chi + \frac{\Delta G_H}{RT}$$

For solubility $\Delta G_H \leq 0$ Attractive

or $\Phi_A \Phi_B \chi \geq 0$ Repulsive

Polystyrene with Polyvinyl-methyl ether





unstable region
metastable region
LCST
single phase
UCST
unstable region
metastable region
Temperature
T_5
T_4
T_3
T_2
T_1
0
1
ϕ_1

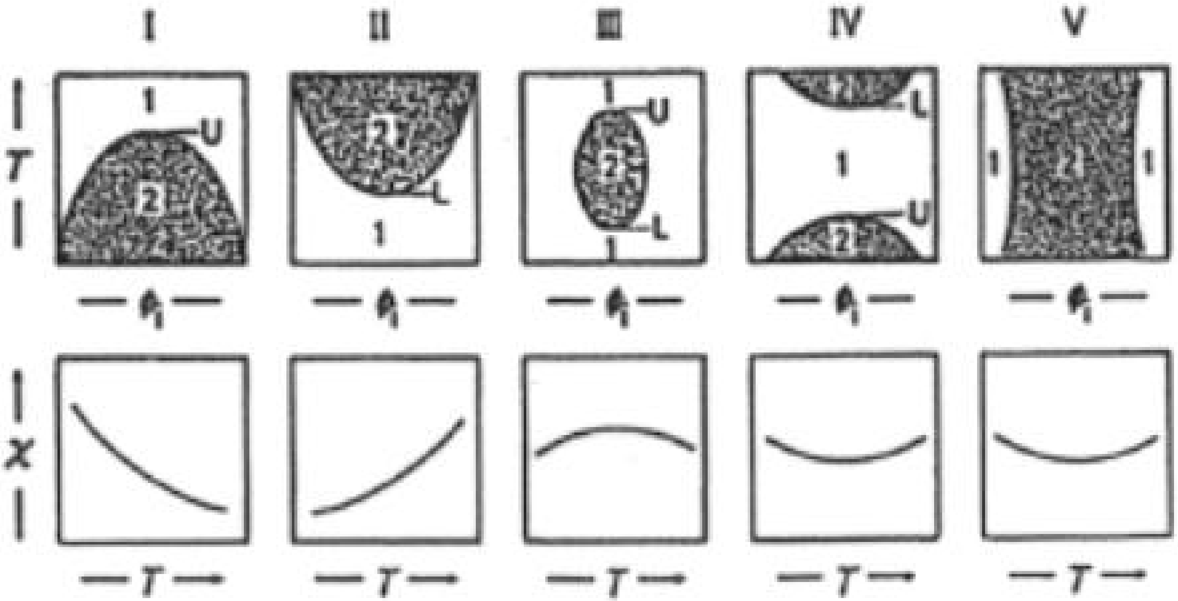


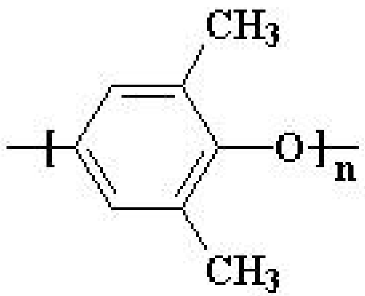
Fig. 6-6 Top: Types of phase diagrams $T = f(\phi_1)$ of fluid systems (polymer solutions or polymer melts) with one-phase (1) or two-phase (2) regions and upper (U) and lower (L) critical solution temperatures. III: Closed miscibility loop; V = hour-glass diagram. Bottom: Temperature dependence of interaction parameters.

Curves are drawn for idealized systems with $X_1 = X_2$ and $\chi \neq f(\phi_2)$. Real systems usually have asymmetric phase diagrams because $X_1 \neq X_2$ and $\chi = f(\phi_2)$. Examples:

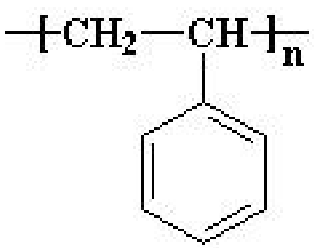
Solutions
I: Poly(styrene)–cyclohexane
II: Poly(ethylene)–hexane (5 bar)
III: Poly(oxyethylene)–water
IV: Poly(styrene)–acetone (low molar mass)
V: Poly(styrene)–acetone (high molar mass)

Blends
Poly(butadiene) (deuterated *vs.* nondeuterated)
Poly(styrene)–poly(vinylmethyl ether)
Poly(methyl methacrylate)–polycarbonate A

Miscible Blend based on weak interactions

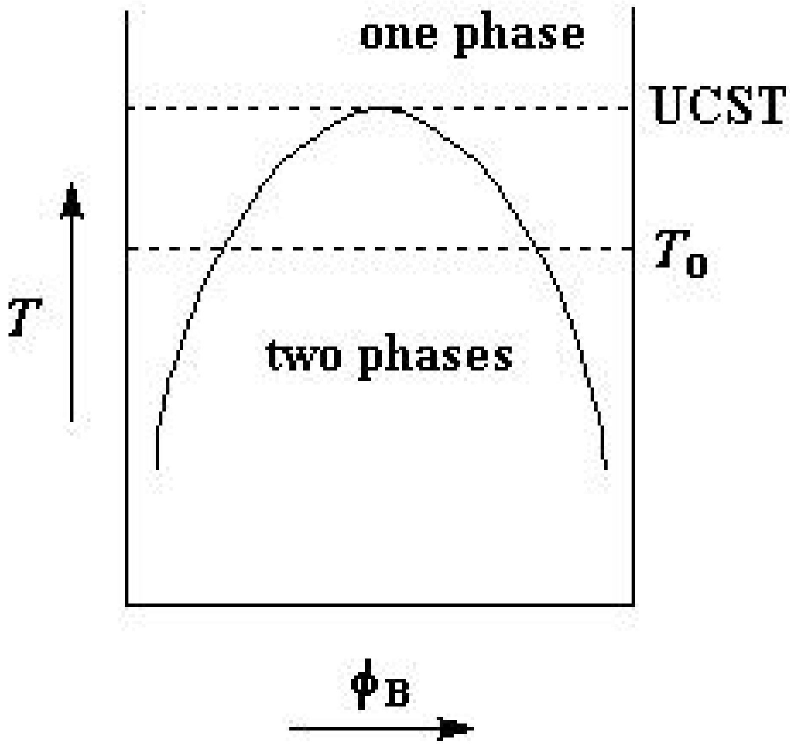


polyphénylène oxyde



polystyrène

Noryl

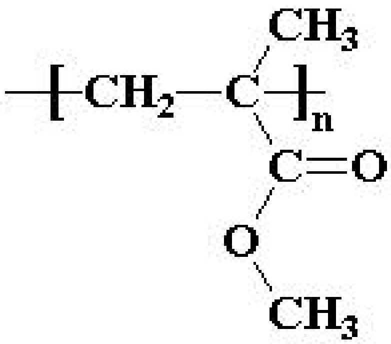


Some others: PMMA-PEO, PVA-PEO, PS-PVME

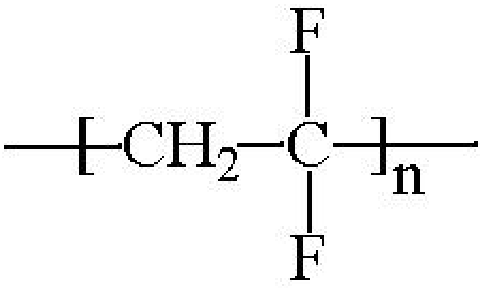
Summary of the Upper Limit of the Critical Values of the Solubility Parameter Difference.

Specific Interactions Involved	Polymer Blend Examples	
Dispersive Forces Only	PBD - PE	≤ 0.1
Dipole - Dipole	PMMA - PEO	0.5
Weak	PVC - BAN	1.0
Weak to Moderate	PC - Polyesters	1.5
Moderate	SAN - PMMA	2.0
Moderate to Strong	Nylon - PEO	2.5
Strong	PVPh - PVAc	3.0
Very Strong	PMAA - PEO	≥ 3.0

Miscible Blend



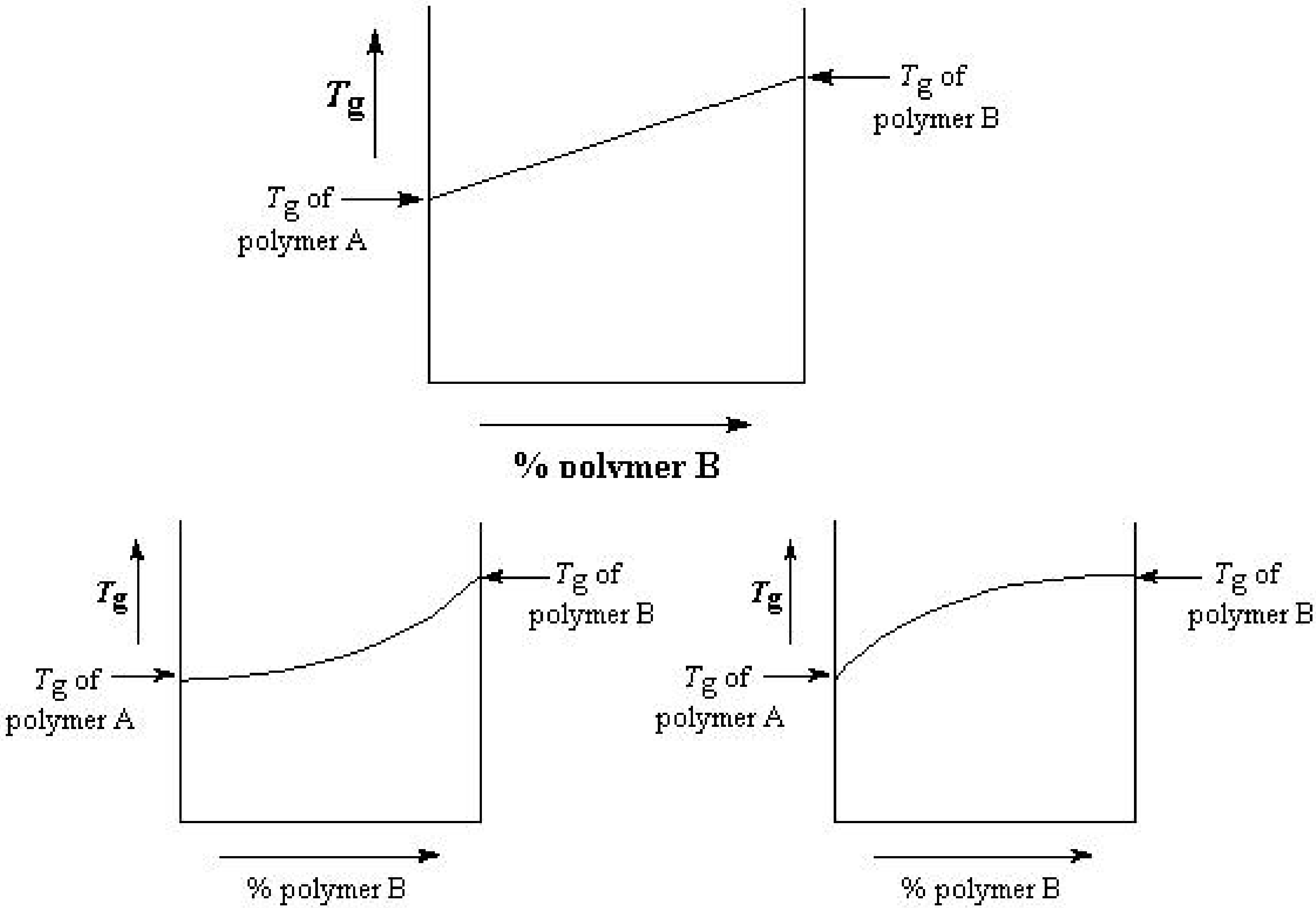
PMMA



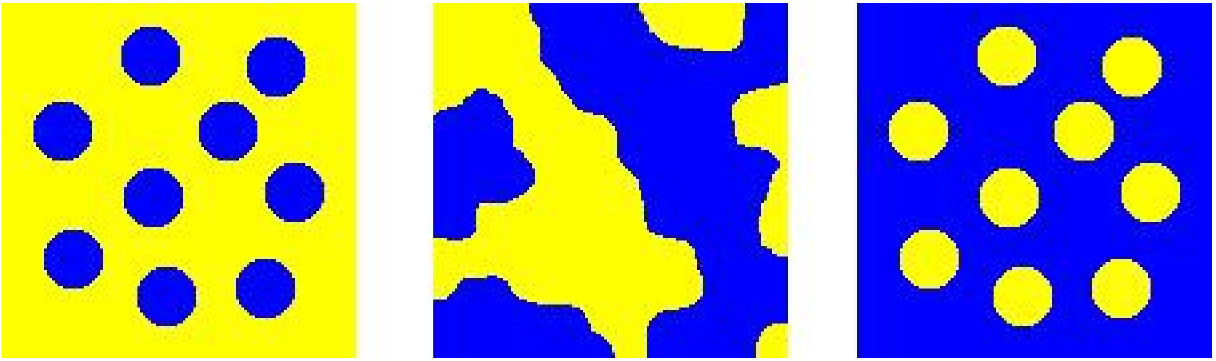
PVF

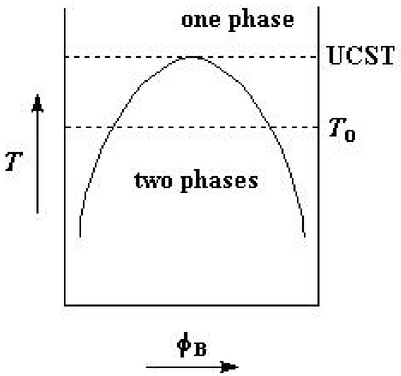
PVF is blended into PMMA to make the latter more resistant to UV

Effects of Blends on Tg



Immiscible Blends

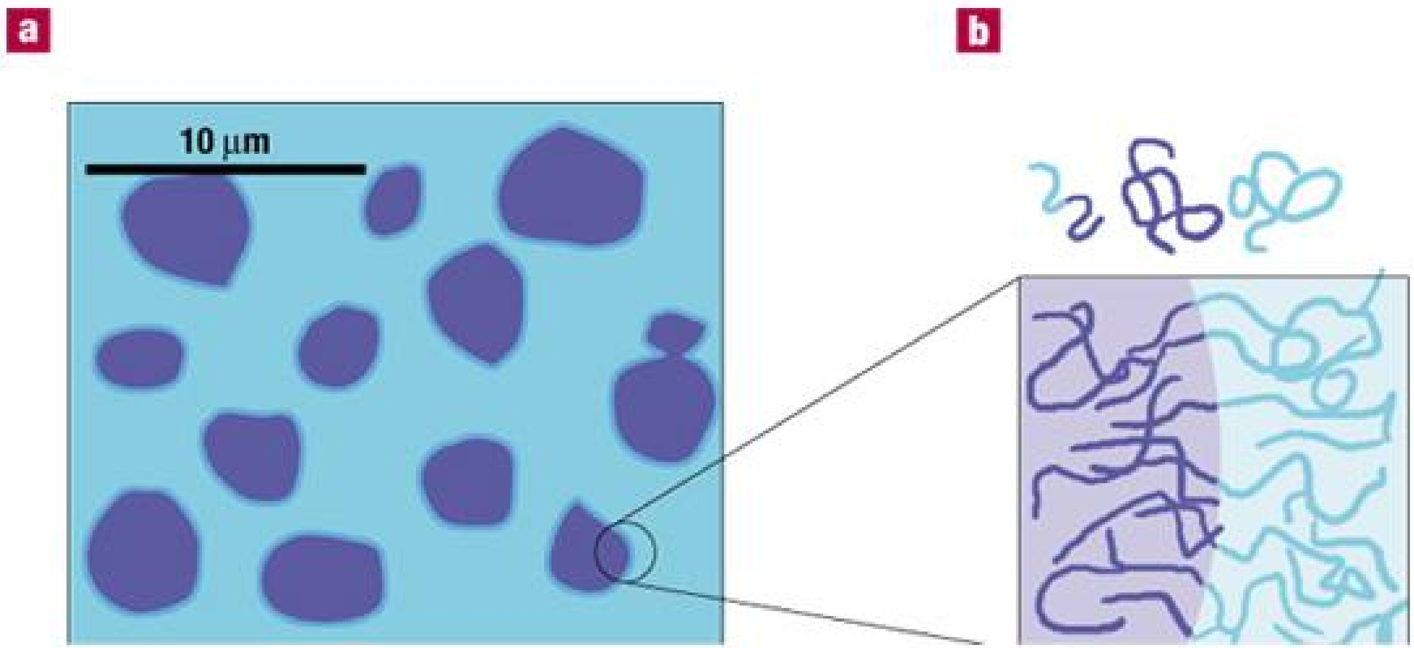




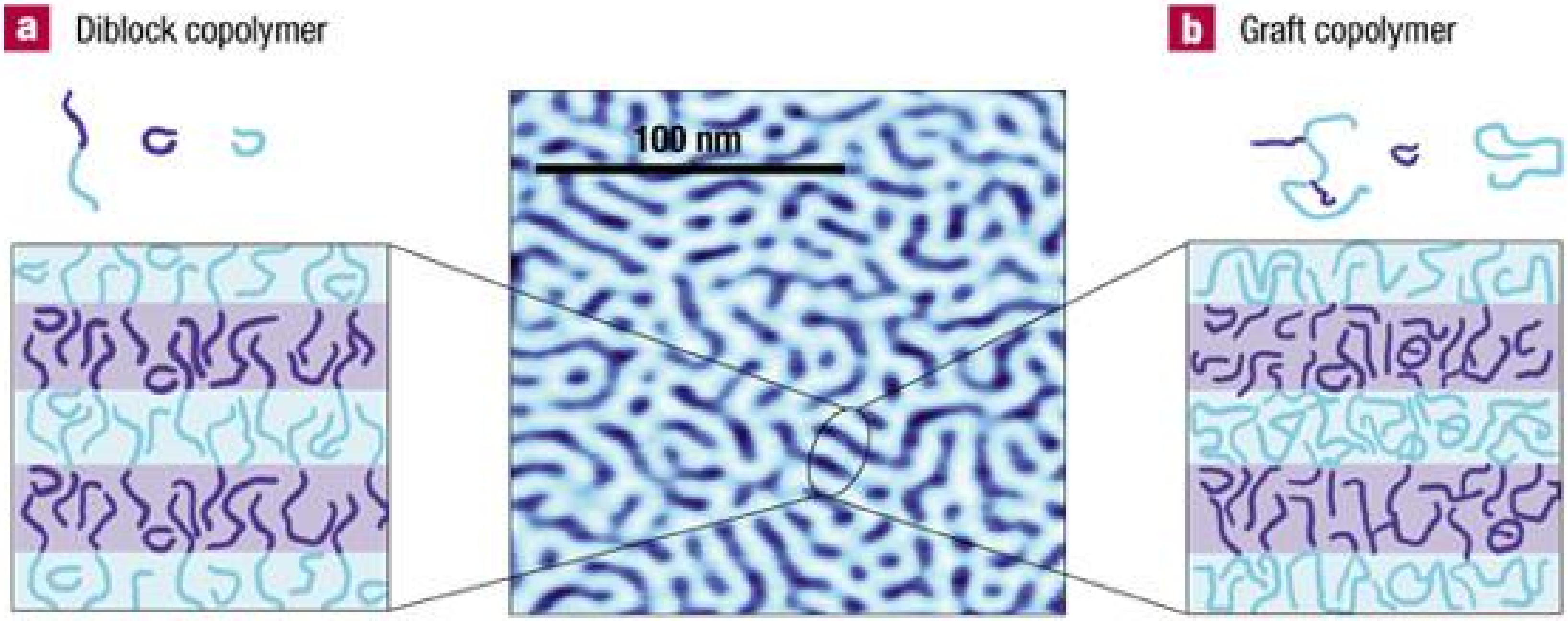
Controlling Phase segregation

- Compositional quenching
- Surfactants
- block polymers
- Compatibilizers

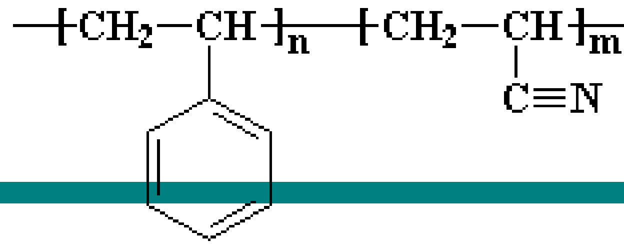
Immiscible Blends



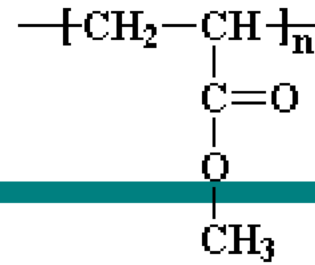
Nature Materials, **2002**, *1*, 8 - 10



a
Diblock copolymer
100 nm
b
Graft copolymer



poly(styrene-*co*-acrylonitrile)



poly(methyl methacrylate)

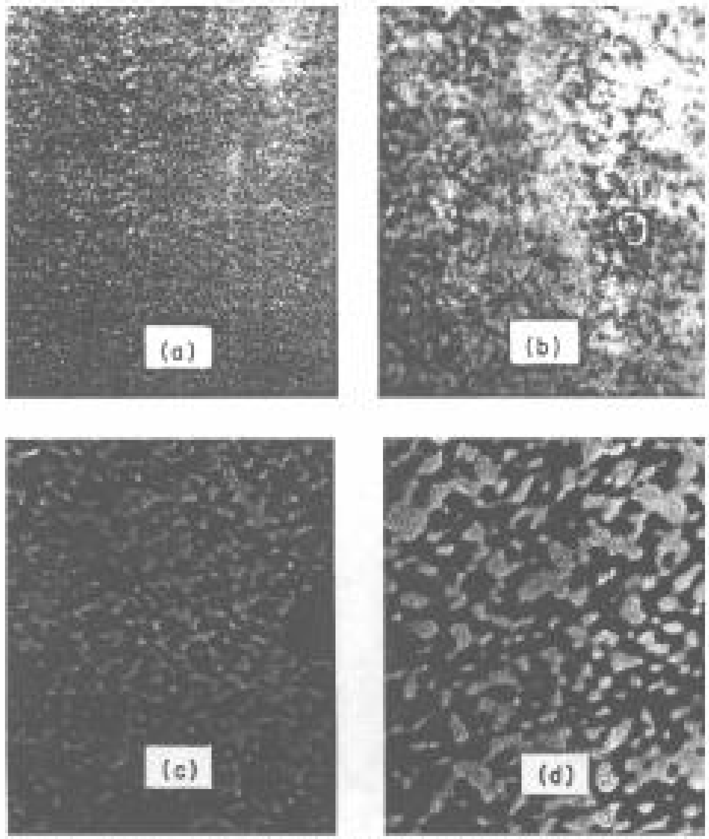
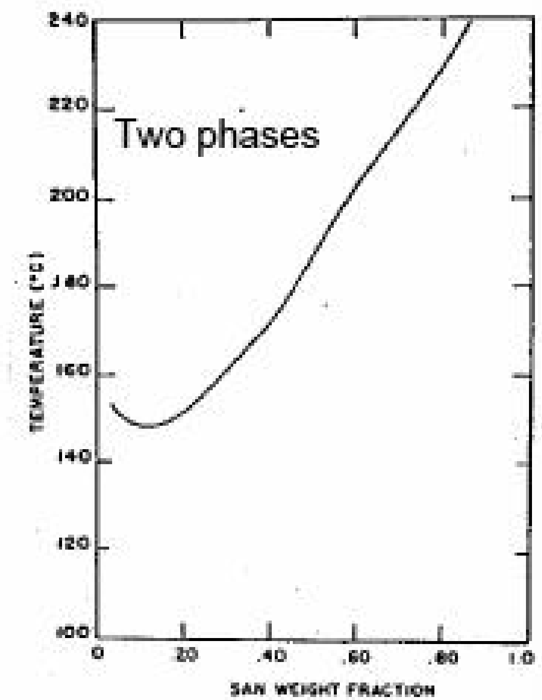


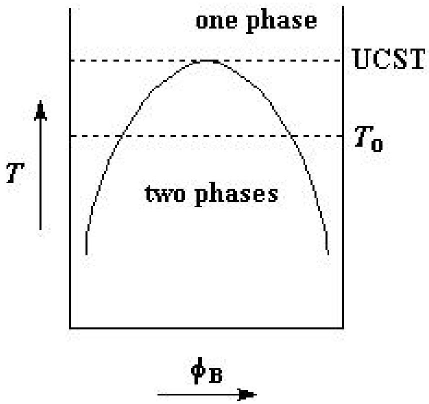
Fig. 28 TEM of stages of phase separation for interconnected structures 25% SAN-75% PMMA; T = 180°C. Time (min): (a) 3.4; (b) 4.4; (c) 13.4; (d) 32.4. The white bar in the figure corresponds to 0.3 μm. (From L. P. McMaster, "Co-polymers, Polyblends, and Composites," Advances in Chemistry Series No. 142, 1975. Copyright by the American Chemical Society.)

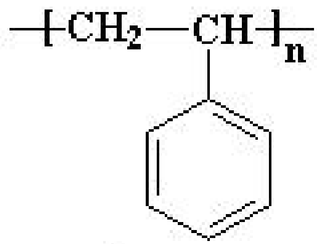


LCST

SAN-PMMA

Using Immiscibility to an Advantage



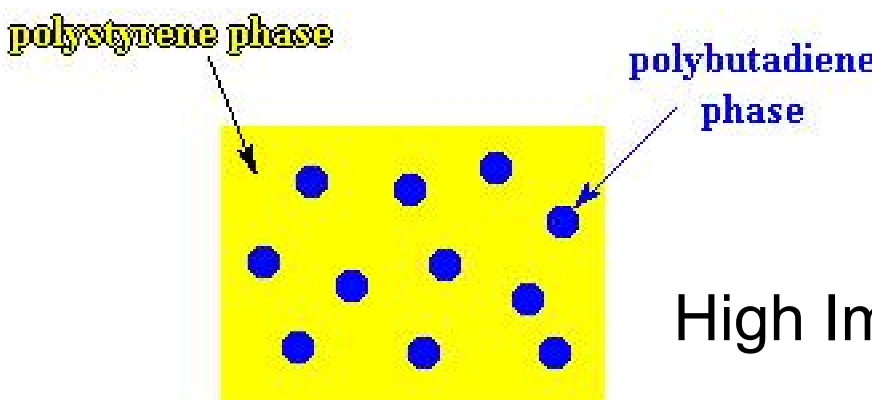


polystyrene

$-\!\!\left[CH_2-CH=CH-CH_2\right]_n\!\!-$

polybutadiene

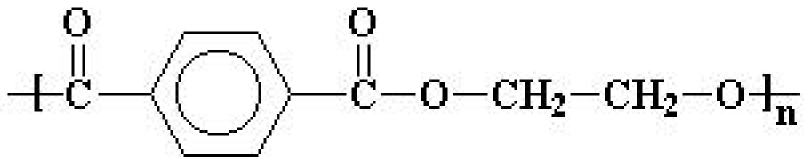
Completely immiscible



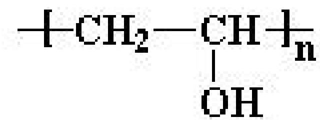
High Impact Polystyrene

Best if the polymers are connected into a *block-copolymer*

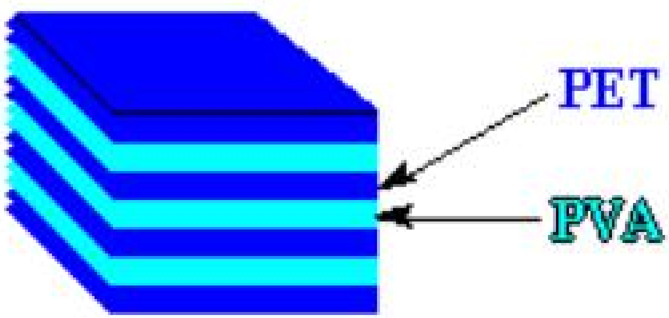
Using Immiscibility to an Advantage



poly(ethylene terephthalate)



poly(vinyl alcohol)

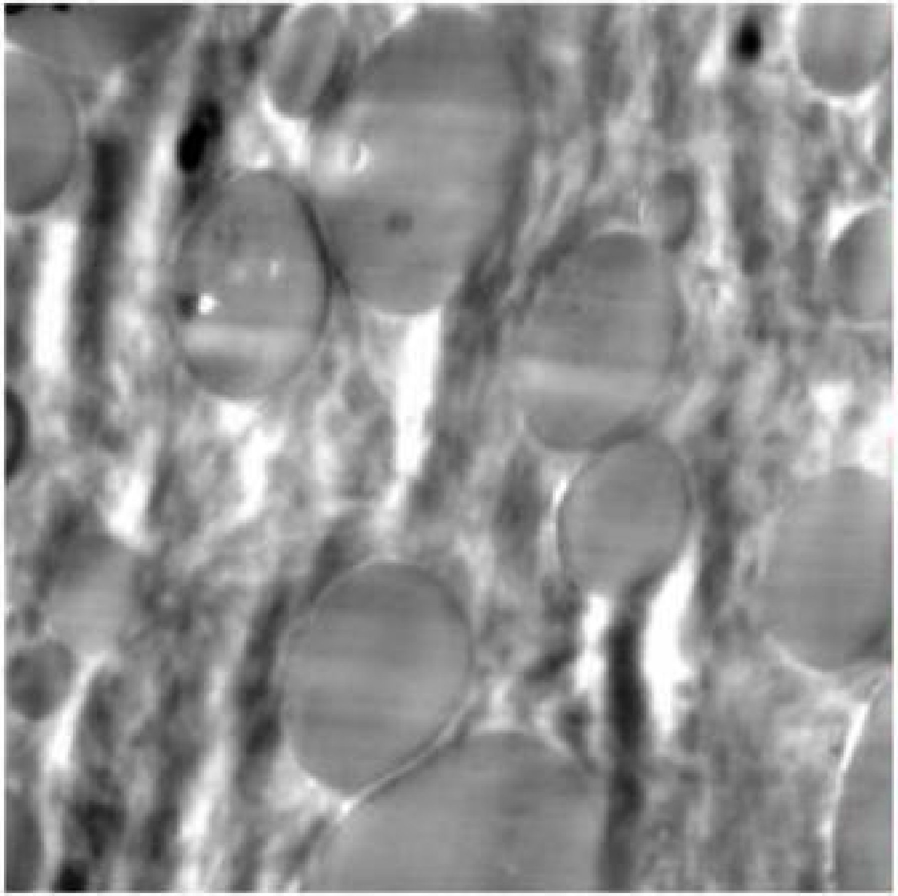


Lining of PET soda bottles

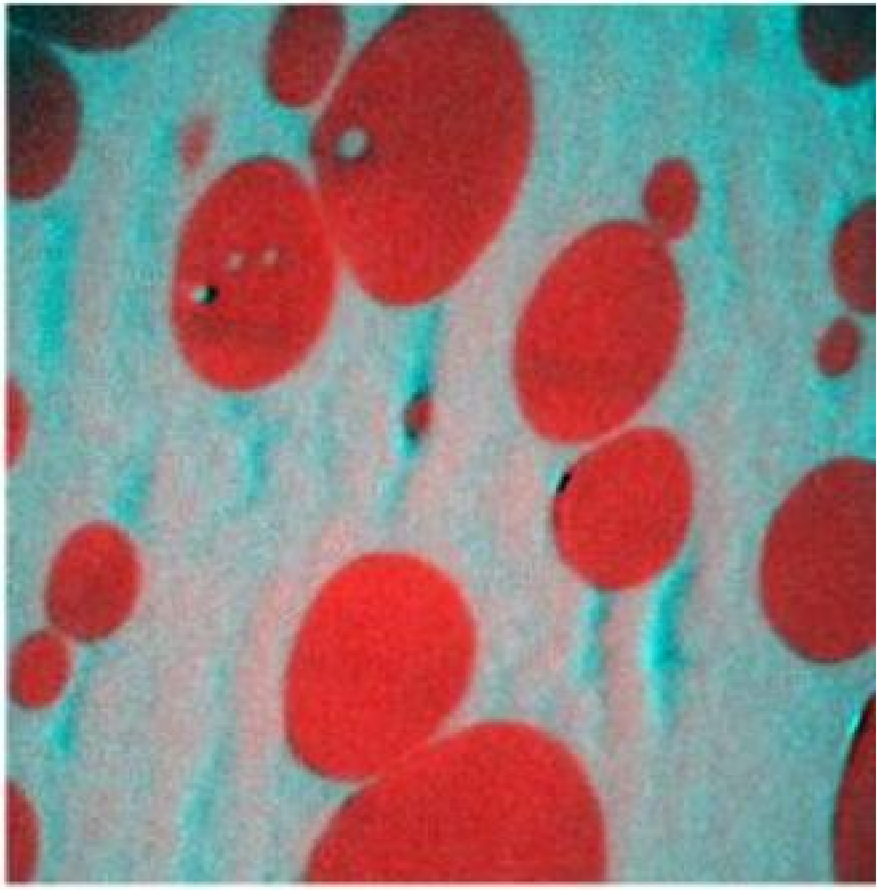
lamellar morphology

EFTEM-investigation of a polyamide-polyphenylenether-blend

Specimen: W. Heckmann, BASF Ludwigshafen



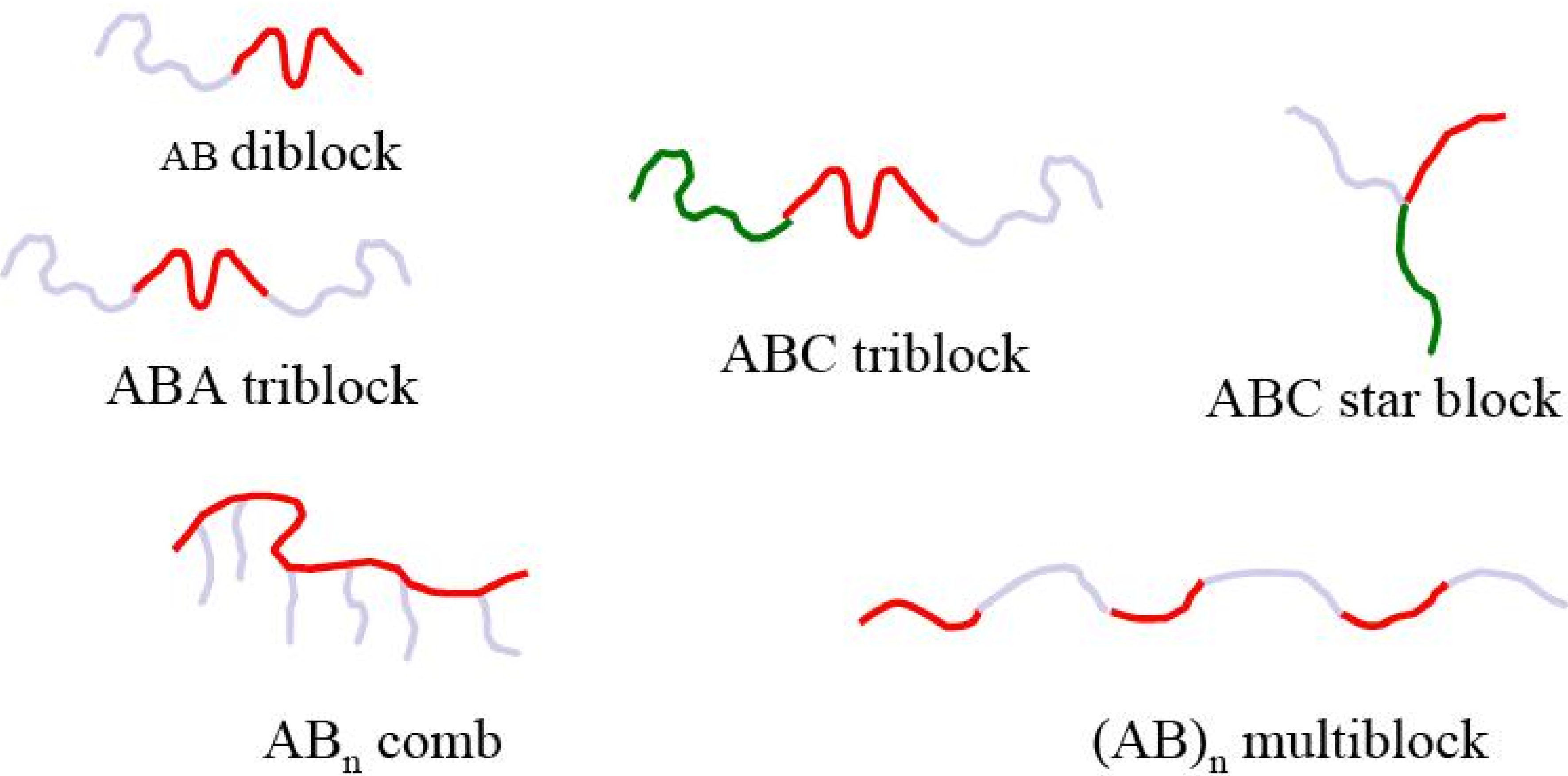
TEM bright field image (zero loss filtered)



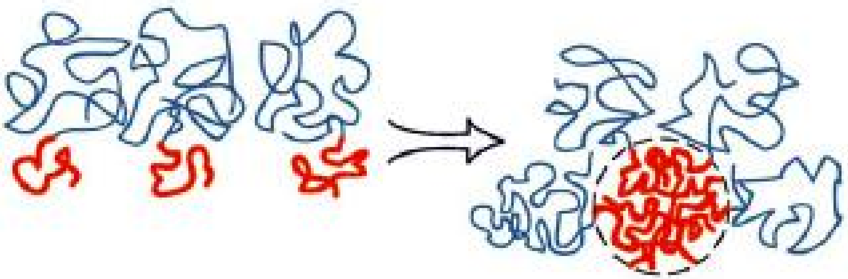
RGB image: red = carbon
blue = nitrogen

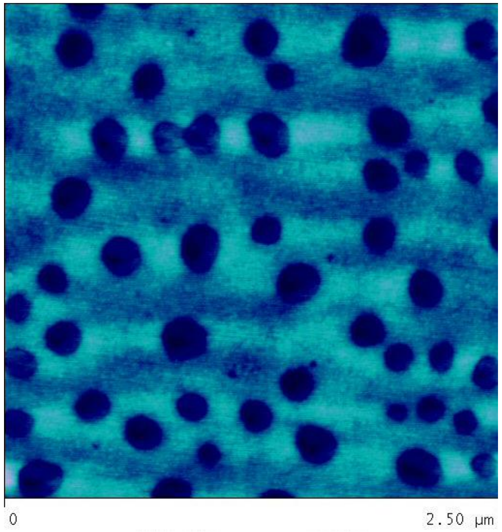
1 µm

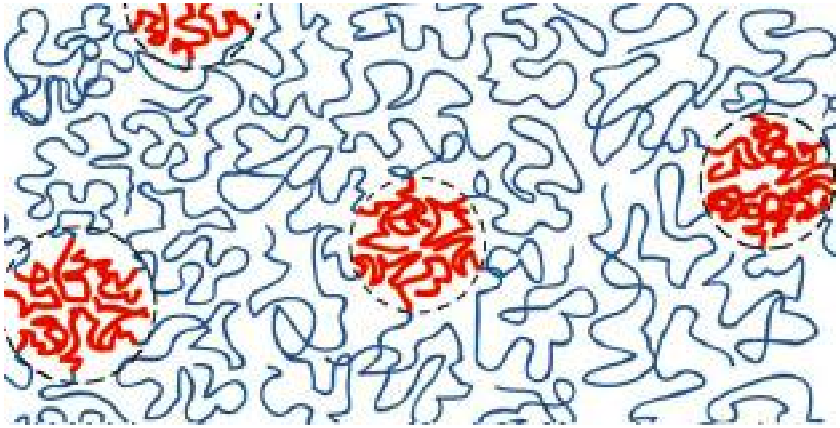
Block Copolymers



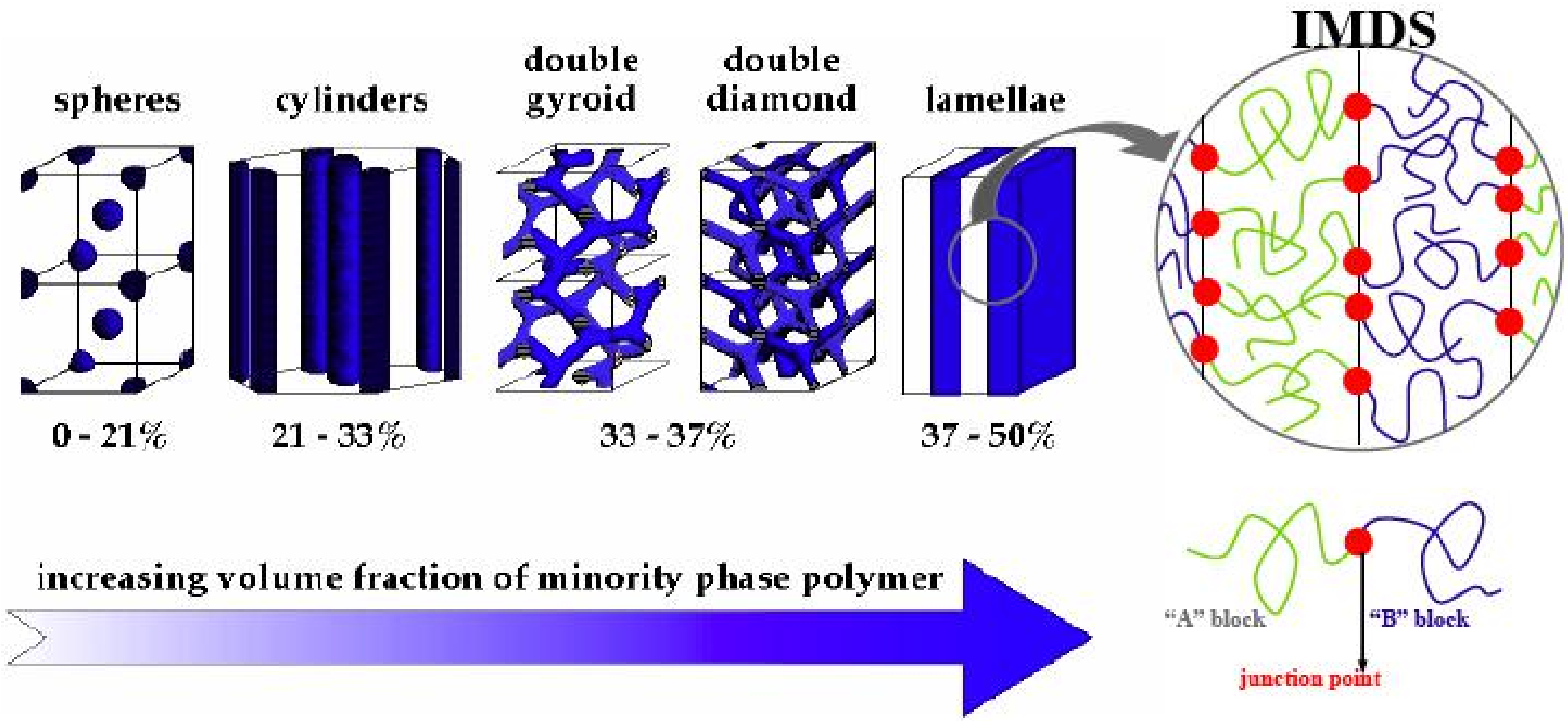
Block Copolymers



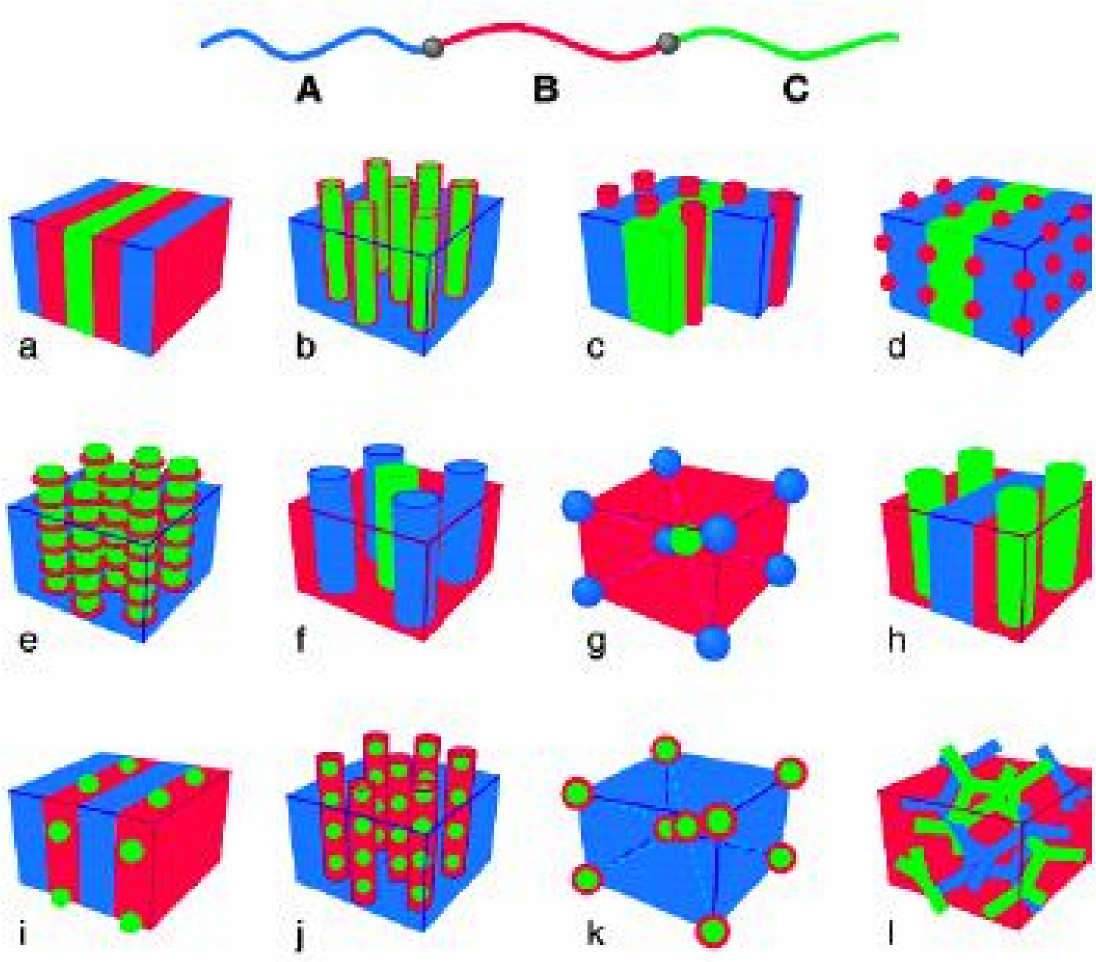




Diblock Copolymers



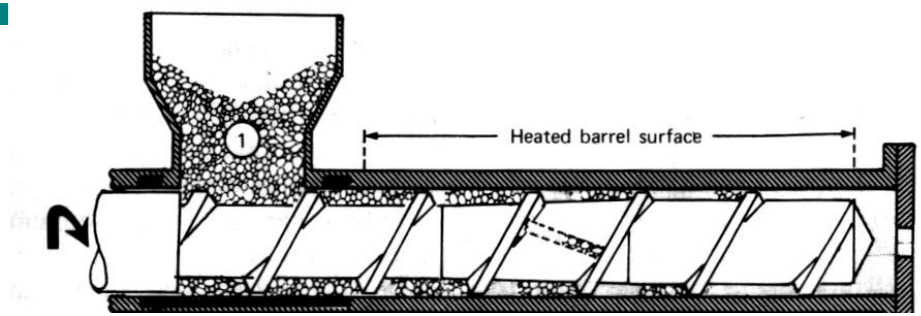
Linear ABC Terpolymer

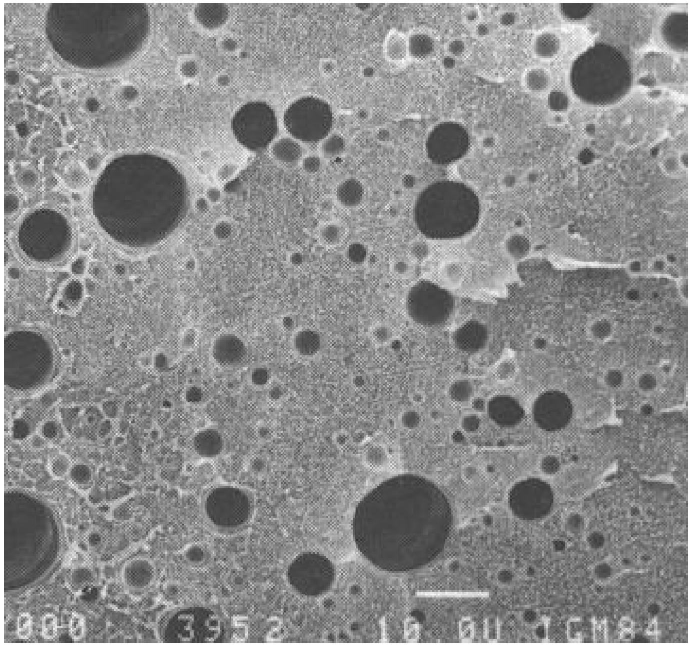


Blends are not easy to make

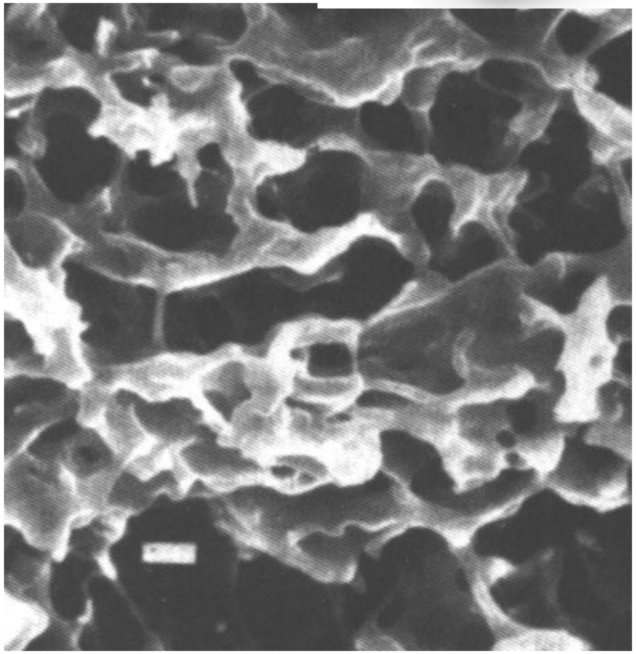
- Blends are not easy to discover-often copolymers are needed to make blend with another homopolymer
- Phase separation gets out of hand in immiscible systems without compatibilizers.
- Some blends must be made with solvent.

Immiscible Blends with 2 homopolymers



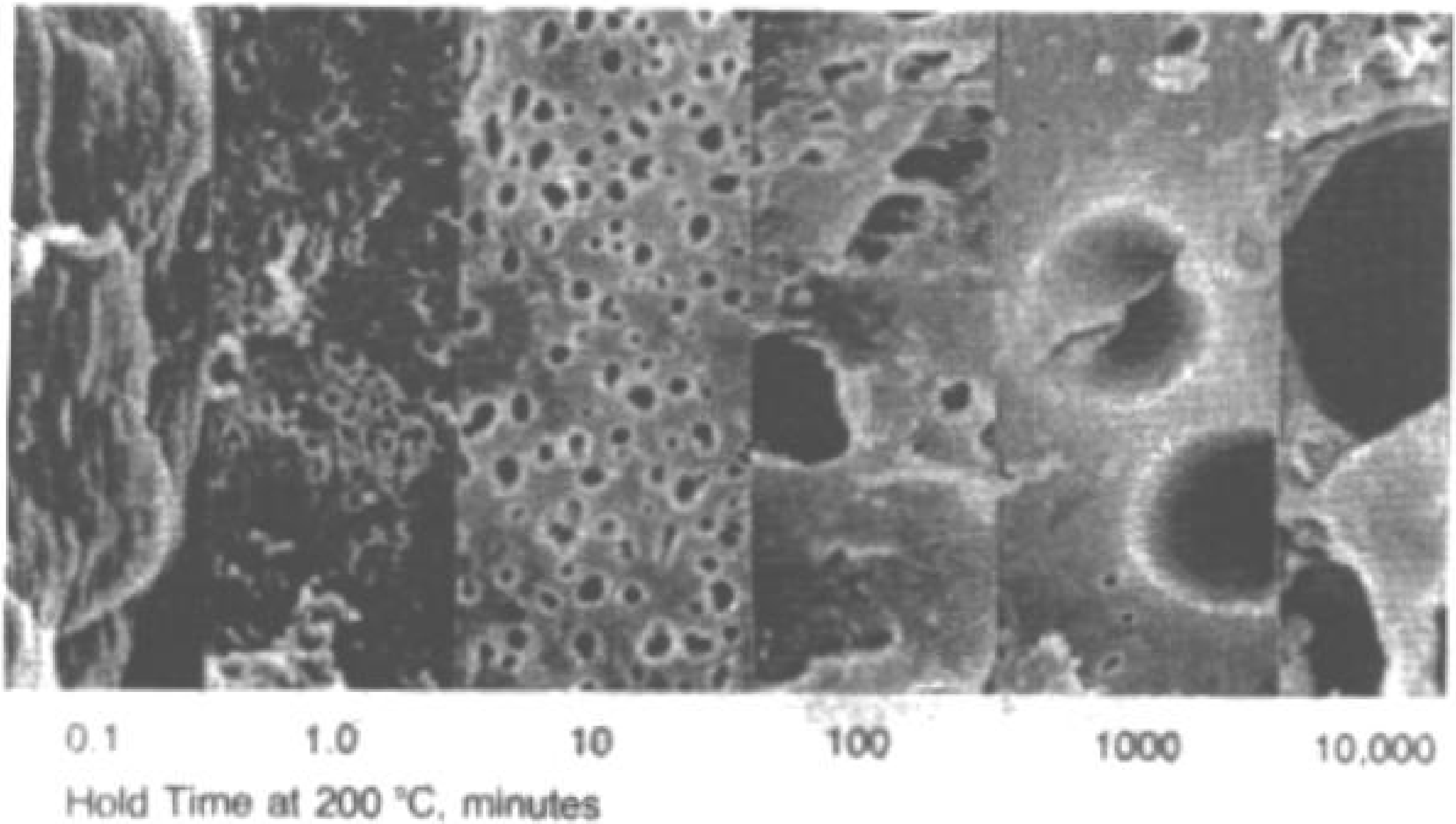


HDPE + 30 wt% Nylon 6

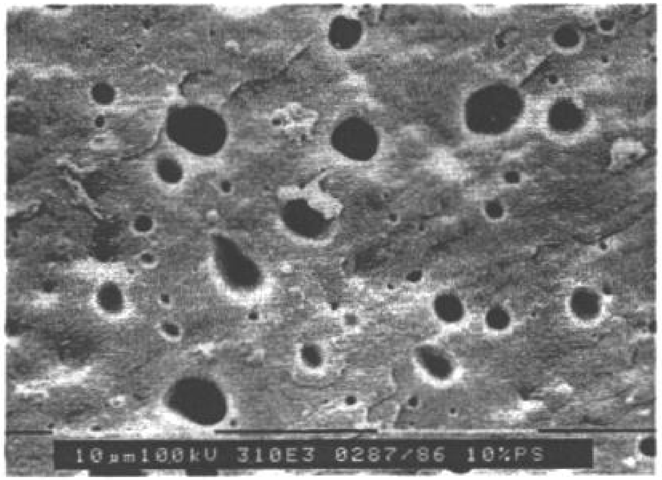


EPR + 30 wt% PP

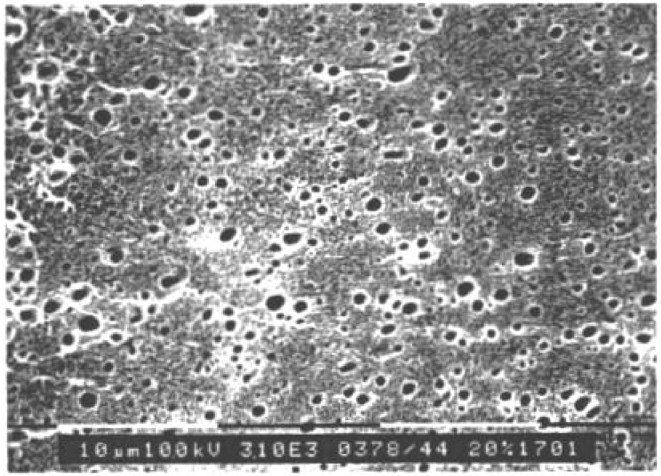
Immiscible Blends with 2 homopolymers can be mixed
But with time coalescence coarsens



Immiscible Blends with 2 homopolymers & compatiblizer



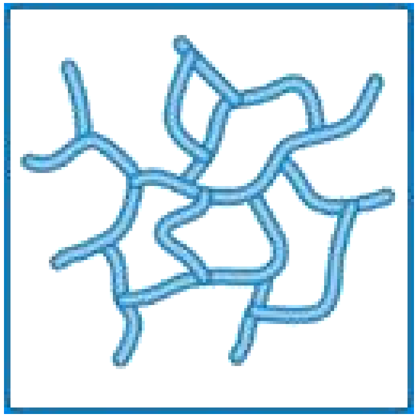
LDPE/PS blend without a compatibilizer



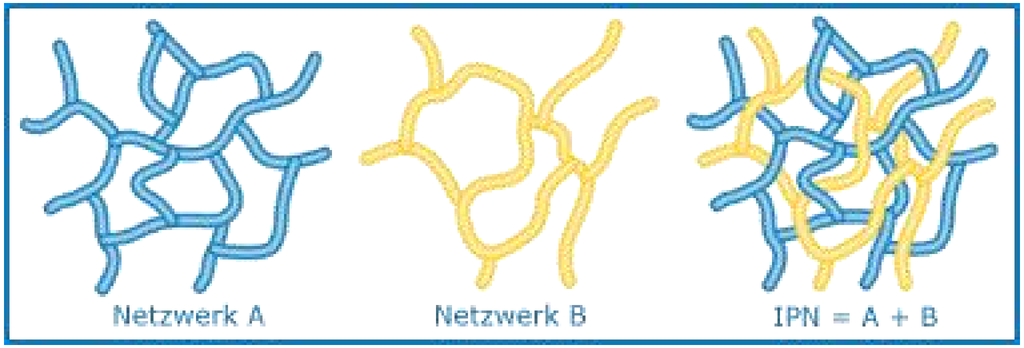
LDPE/PS blend with a compatibilizer

Polymer IPNS

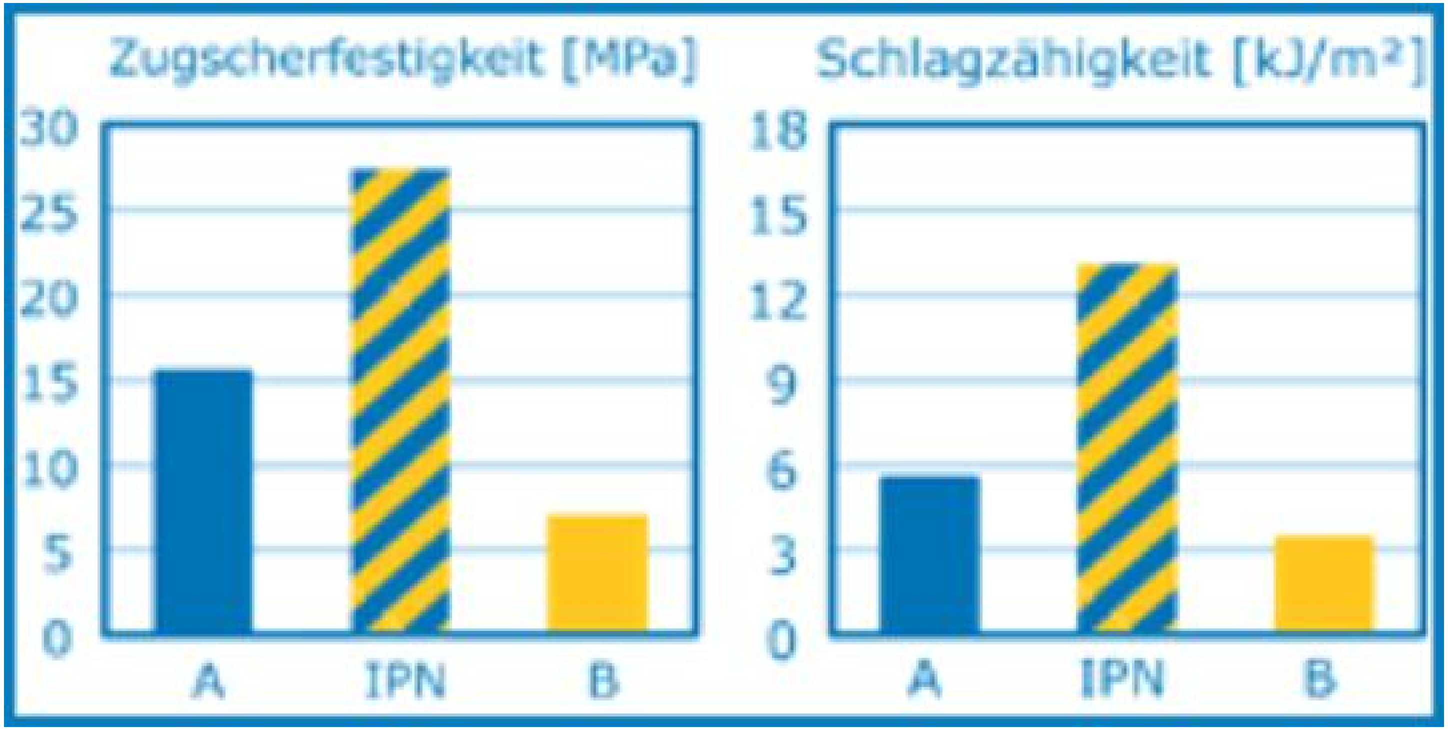
Conventional epoxy Networks



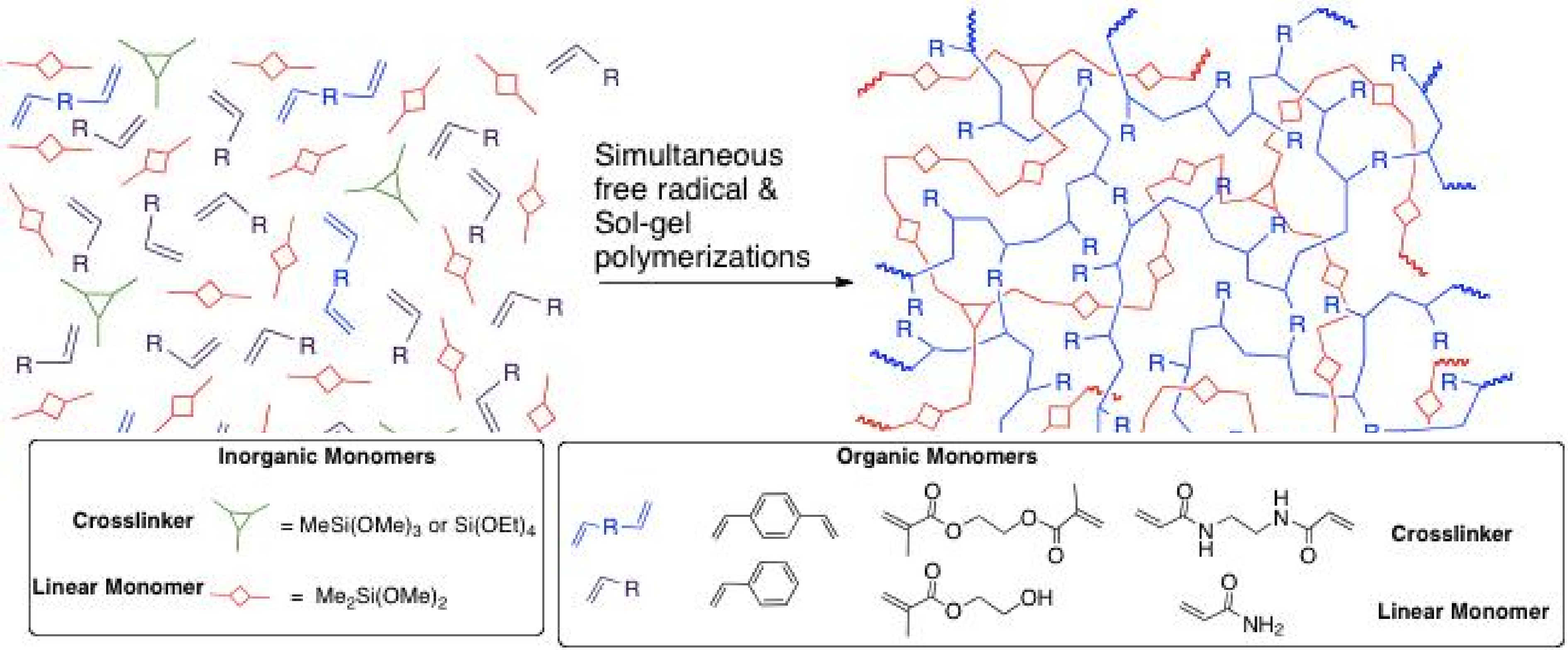
Interpenetrating Networks



Polymer IPNS

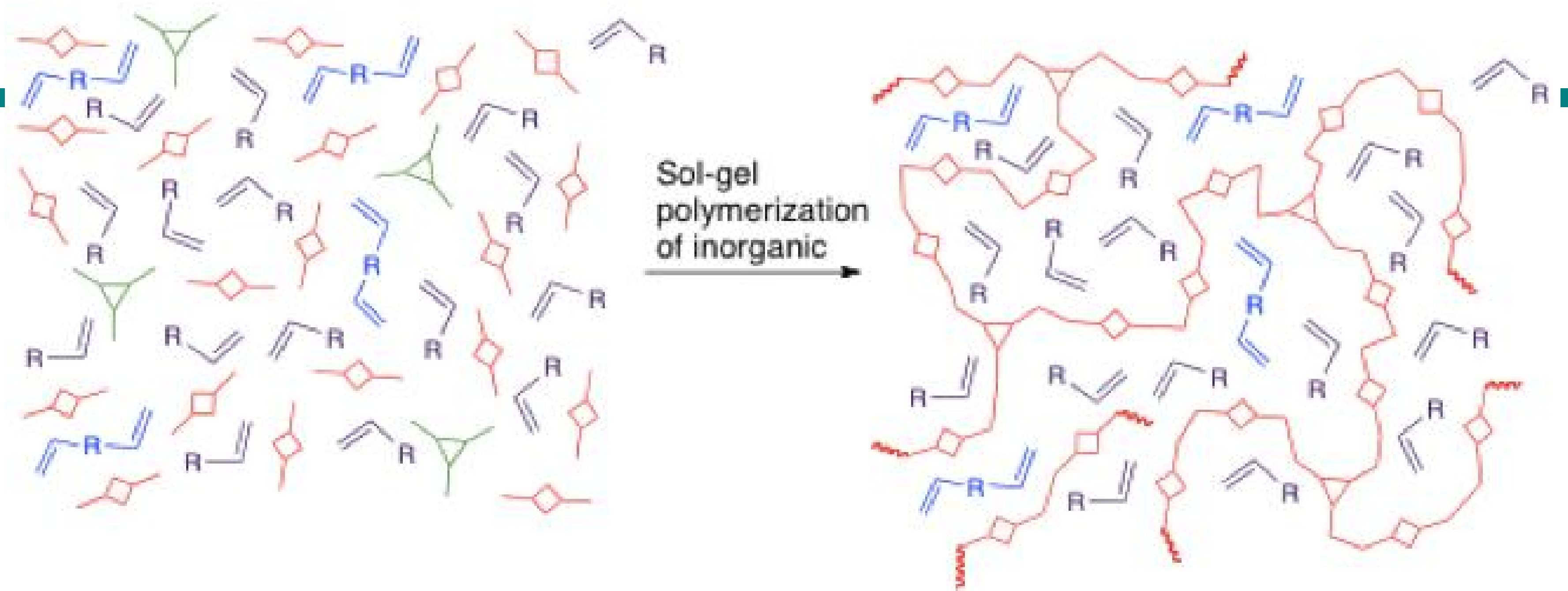


Simultaneous Interpenetrating Network Polymers

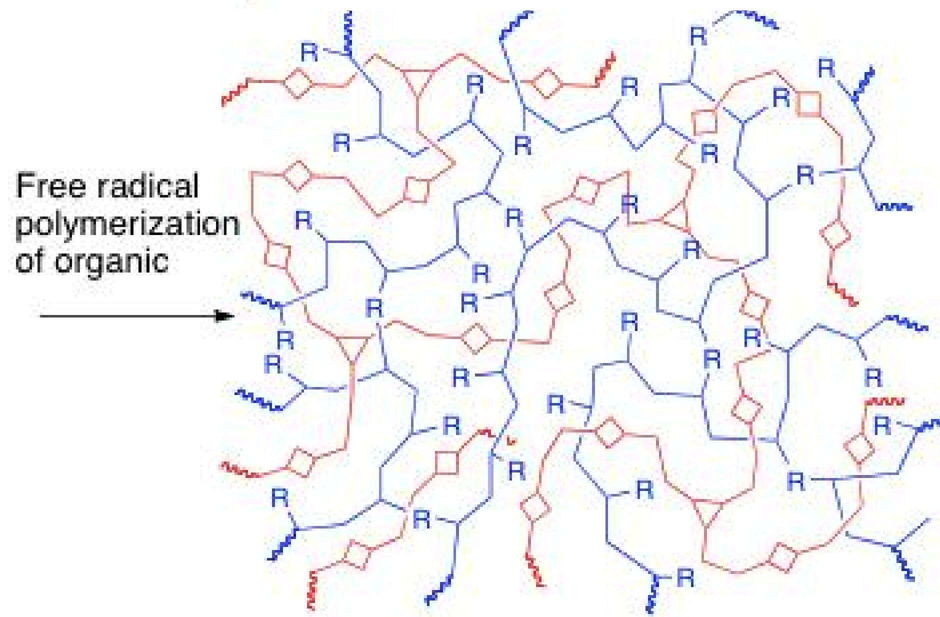


- **The two networks assembled at the same time**
- **Phase separation of hard colloidal particles would ruin IPN**

Sequential Interpenetrating Networks



- **The two networks are not assembled at the same time**
- **In this case the inorganic network is assembled before the organic**



Composites

- The essence of the concept of composites is that the load is applied over a large surface area of the matrix. Matrix then transfers the load to the reinforcement, which being stiffer, increases the strength of the composite. It is important to note that there are many matrix materials and even more fiber types, which can be combined in countless ways to produce just the desired properties.

- About 90% of all composites produced are comprised of glass fiber and either polyester or vinylester resin. Composites are broadly known as reinforced plastics.

Composites

A composite is:

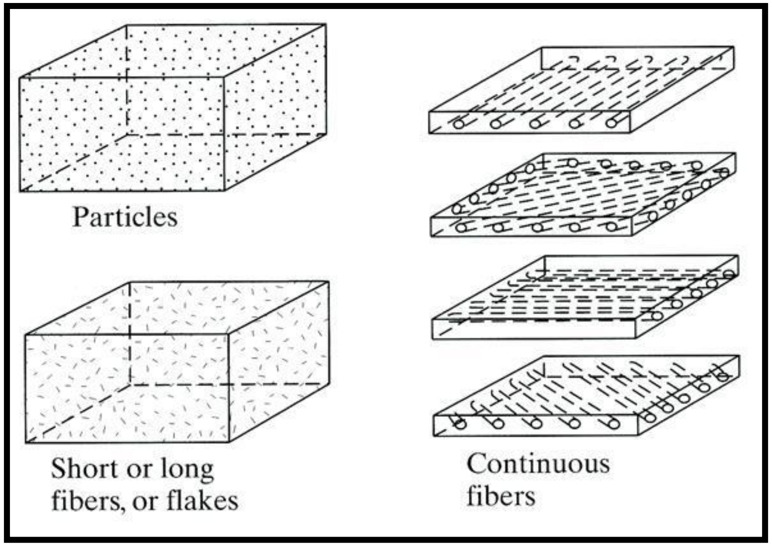
- Two or more distinct materials

- that interface on a macroscopic scale

- to achieve best properties not possessed by any one acting alone.

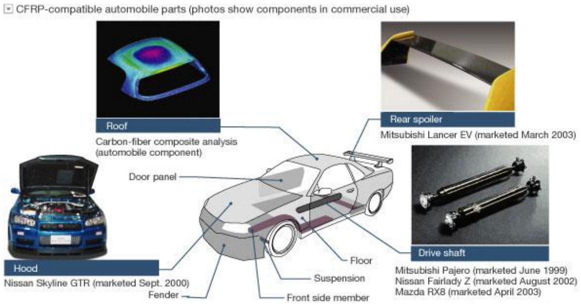
Composites – *Polymer Matrix*

- Polymer matrix composites (PMC) and fiber reinforced plastics (FRP) are referred to as *Reinforced Plastics.* Common fibers used are glass (GFRP), graphite (CFRP), boron, and aramids (Kevlar). These fibers have *high specific strength* (strength-to-weight ratio) and *specific stiffness* (stiffness-to-weight ratio)



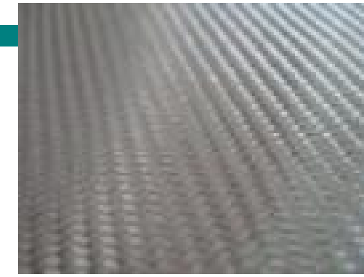
Matrix materials are usually thermoplastics or thermosets; polyester, epoxy (80% of reinforced plastics), fluorocarbon, silicon, phenolic.

Composites are all around us



Composites have a long use in military

A composite has three elements



Reinforcement:
Makes up the ‘skeleton’ which provides mechanical strength

Matrix:
Binds the reinforcing fibers and distributes the load

PETE HDPE PS
PP PVC
PEEK

Additives / Fillers:
Provide special attributes not in the base materials

Wet Out Sizing
Talc Chalk
Flowablility Fire Retardant

What makes composites attractive

Reduced weight

High specific strength and stiffness

Good fatigue strength

Energy absorbing

Corrosion resistance

High resistance to chemicals and weathering (UV)

Functional integration

Design flexibility

Durability:

Main limitations of composites

Initial expense

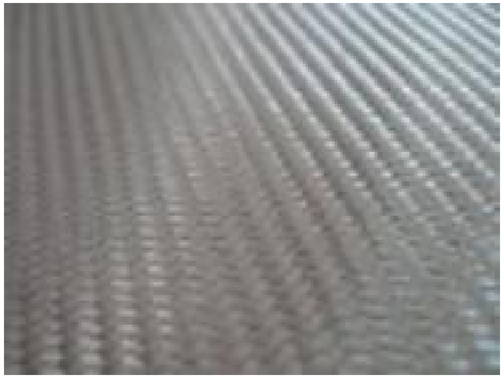
Impact strength

High temperature

Reinforcement Forms

Woven

perpendicular warp and weft yarns



Glass fiber
E-glass
C-glass
S-glass

Carbon (PAN or Pitch)

Polymer:
Kevlar
Nomex

Ceramic: Si-C, Si_3N_4, or BN

Typical properties of fibers

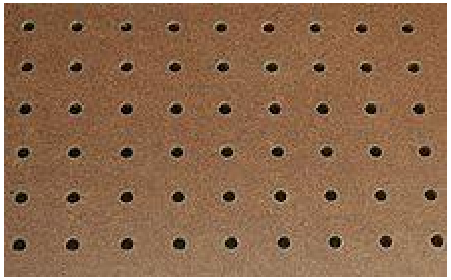
Fibre type	Fibre diameter μm	Specific gravity	Tensile modulus GPa	Tensile strength MPa	Failure strain %
E-glass (glass)	10	2.54	73	2,450	4.8
T-300 (carbon)	7	1.76	228	3,200	1.4
Kevlar-49 (aramid)	11.9	1.45	131	3,620	2.8

Wood-Polymer Composites

Oriented strand board

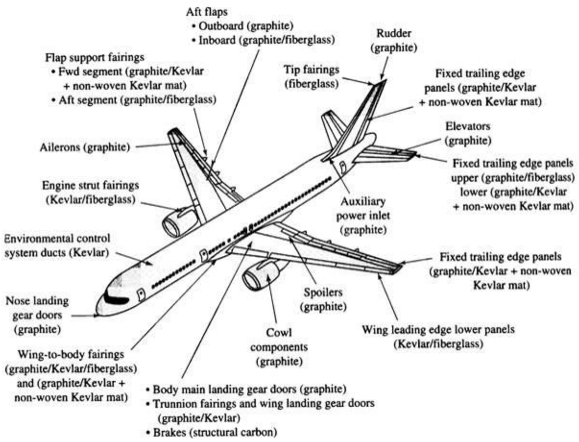
Particle Board

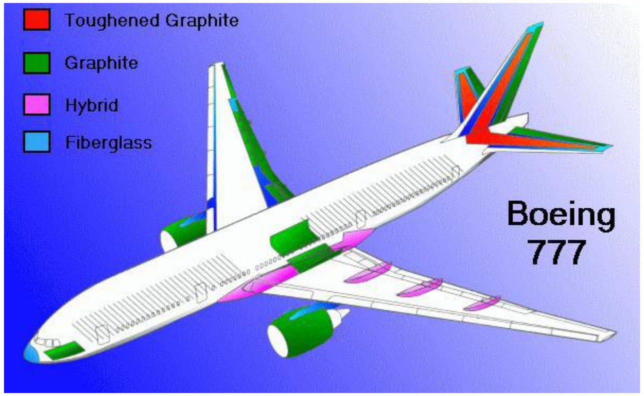
Masonite

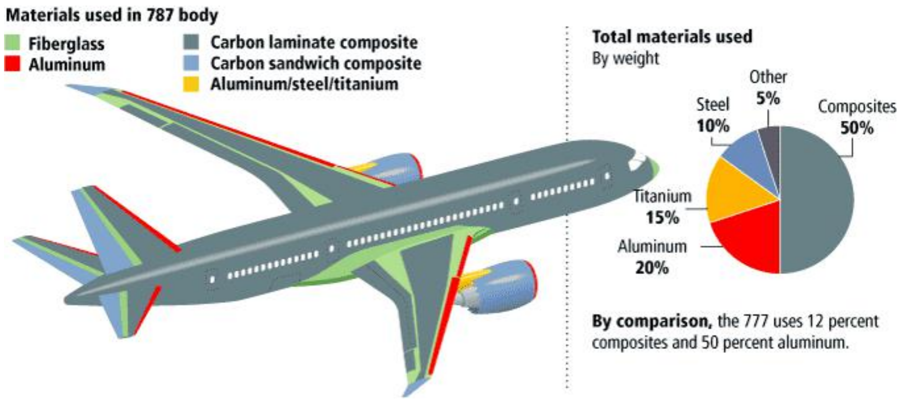


Pegboard

Composites are used in Commercial Aircraft





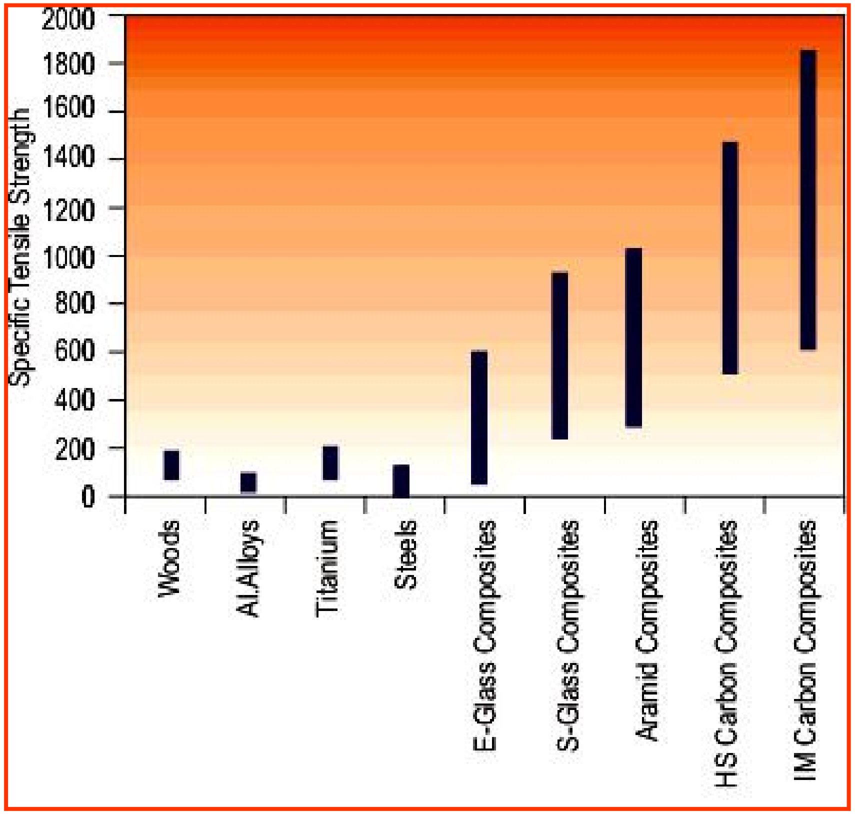


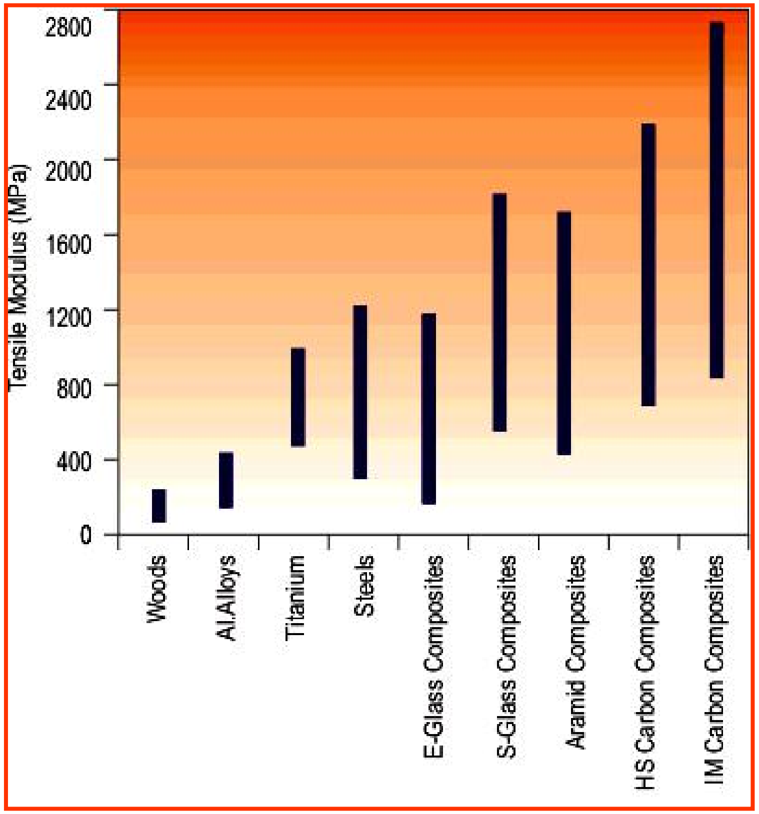
Application of Composites in Aircraft Industry

Boeing 777	Boeing 787/Dreamliner
Launched in 2000	*To be launched in 2007*
11% composites	50% composites
70% aluminum	20% aluminum
7% titanium	15% titanium
11% steel	10% steel
1% other	5% other

20% more fuel efficiency and 35,000 lbs. lighter

Advantages of Composites





Advantages of Composites

- **Design flexibility**

Composites have an advantage over other materials because they can be molded into complex shapes at relatively low cost. This gives designers the freedom to create any shape or configuration. Boats are a good example of the success of composites.

- **Corrosion Resistance**

Composites products provide long-term resistance to severe chemical and temperature environments. Composites are the material of choice for outdoor exposure, chemical handling applications, and severe environment service.

Advantages of Composites

- **Low Relative Investment**

One reason the composites industry has been successful is because of the low relative investment in setting-up a composites manufacturing facility. This has resulted in many creative and innovative companies in the field.

- **Durability**

Composite products and structures have an exceedingly long life span. Coupled with low maintenance requirements, the longevity of composites is a benefit in critical applications. In a half-century of composites development, well-designed composite structures have yet to wear out.

Disadvantages of Composites

- **Composites are heterogeneous**

Properties in composites vary from point to point in the material. Most engineering structural materials are homogeneous.

- **Composites are highly anisotropic**

The strength in composites vary as the direction along which we measure changes (most engineering structural materials are isotropic). As a result, all other properties such as, stiffness, thermal expansion, thermal and electrical conductivity and creep resistance are also anisotropic. The relationship between stress and strain (force and deformation) is much more complicated than in isotropic materials.

Disadvantages of Composites

Composites materials are difficult to inspect with conventional ultrasonic, eddy current and visual NDI methods such as radiography.

American Airlines Flight 587, broke apart over New York on Nov. 12, 2001 (265 people died). Airbus A300's 27-foot-high tail fin tore off. Much of the tail fin, including the so-called tongues that fit in grooves on the fuselage and connect the tail to the jet, were made of a graphite composite. The plane crashed because of damage at the base of the tail that had gone undetected despite routine nondestructive testing and visual inspections.

HOMEWORK

- Find **3 types of polymer blends** and **3 types of polymer-based composites** and briefly state their properties and applications.